

49

Nervous Systems

KEY CONCEPTS

- 49.1** Nervous systems consist of circuits of neurons and supporting cells
- 49.2** The vertebrate brain is regionally specialized
- 49.3** The cerebral cortex controls voluntary movement and cognitive functions
- 49.4** Changes in synaptic connections underlie memory and learning
- 49.5** Many nervous system disorders can be explained in molecular terms

▲ **Figure 49.1** How do scientists identify individual neurons in the brain?

Command and Control Center

What happens in your brain when you solve a math problem or listen to music? Answering such a question was for a long time nearly unimaginable. The human brain contains an estimated 10^{11} (100 billion) neurons. Interconnecting these brain cells are circuits more complex than those of even the most powerful supercomputers. However, thanks in part to several exciting new technologies, scientists have begun to explore the cellular mechanisms that underlie thought and emotion.

One breakthrough came with the development of powerful imaging techniques that reveal activity in the working brain. Researchers can monitor multiple areas of the human brain while a subject is performing various tasks, such as speaking, looking at pictures, or forming a mental image of a person's face. They can use these techniques to look for a correlation between a particular task and activity in specific brain areas.

A more recent advance in exploring the brain relies on a method for expressing random combinations of colored proteins in brain cells—such that each cell shows up in a different color. The result is a “rainbow” like the one in **Figure 49.1**, which highlights neurons in the brain of a mouse. In this image, each neuron expresses one of more than 90 different color combinations of four fluorescent proteins.

Using the brainbow technology, neuroscientists hope to develop detailed maps of the connections that transfer information between particular regions of the brain.

In this chapter, we'll discuss the organization and evolution of animal nervous systems, exploring how groups of neurons function in specialized circuits dedicated to specific tasks. Next we'll focus on specialization in regions of the vertebrate brain. We'll then turn to the ways in which brain activity makes information storage and organization possible. Finally, we'll consider several disorders of the nervous system that are the subject of intense research today.

CONCEPT 49.1

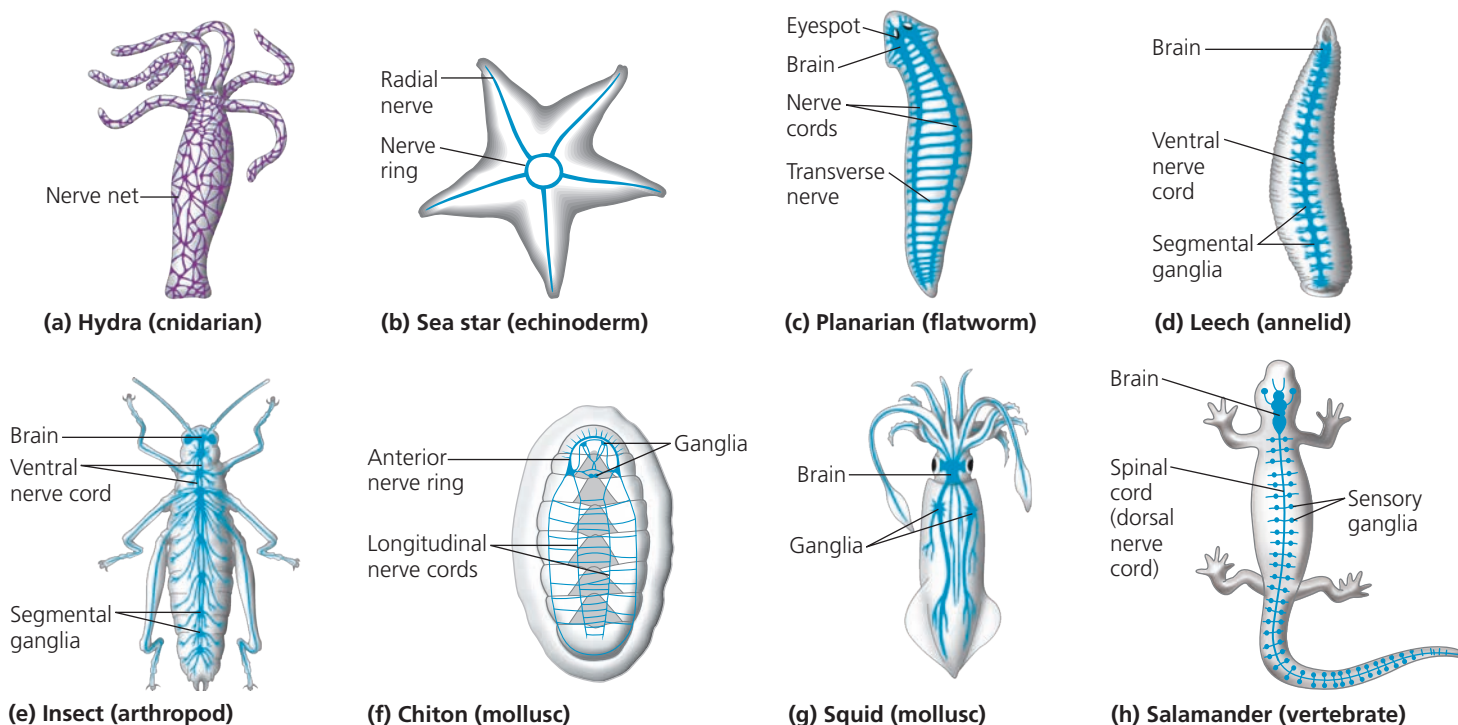
Nervous systems consist of circuits of neurons and supporting cells

The ability to sense and react originated billions of years ago in prokaryotes, enhancing survival and reproductive success in changing environments. Later in evolution, modification of simple recognition and response processes provided a basis for communication between cells in an animal body. By the time of the Cambrian explosion more than 500 million years ago (see Chapter 32), specialized systems of neurons had appeared that enable animals to sense their surroundings and respond rapidly.

Hydras, jellies, and other cnidarians are the simplest animals with nervous systems. In most cnidarians, interconnected neurons form a diffuse *nerve net* (**Figure 49.2a**), which controls the contraction and expansion of the gastrovascular cavity. In more complex animals, the axons of multiple neurons are often bundled together, forming **nerves**. These fibrous structures channel and organize information flow along specific routes through the nervous system. For example, sea stars have a set of radial nerves connecting to a central nerve ring (**Figure 49.2b**). Within each arm of a sea star, the radial nerve is linked to a nerve net from which it receives input and to which it sends signals that control muscle contraction.

Animals that have elongated, bilaterally symmetrical bodies have even more specialized nervous systems. The organization of neurons in such animals reflects *cephalization*, an evolutionary trend toward a clustering of sensory neurons and interneurons at the anterior (front) end of the body. These anterior neurons communicate with cells elsewhere in the body, including neurons located in one or more nerve cords extending toward the posterior (rear) end.

As you learned in Concept 48.1, in many animals neurons that carry out integration form a **central nervous system (CNS)**, and neurons that carry information into and out of the CNS form a **peripheral nervous system (PNS)**. In nonsegmented worms, such as the planarian in **Figure 49.2c**, a small brain and longitudinal nerve cords constitute the simplest



▲ **Figure 49.2 Nervous system organization.** (a) A hydra contains individual neurons (purple) organized in a diffuse nerve net. (b–h) Animals with more sophisticated nervous systems contain groups of neurons (blue) organized into nerves and often ganglia and a brain.

clearly defined CNS. In some nonsegmented worms, the entire nervous system is constructed from only a small number of cells, as in the case of the nematode *Caenorhabditis elegans*. In this species, an adult worm (hermaphrodite) has exactly 302 neurons, no more and no fewer. More complex invertebrates, such as segmented worms (annelids; **Figure 49.2d**) and arthropods (**Figure 49.2e**), have many more neurons. Their behavior is regulated by more complicated brains and by ventral nerve cords containing **ganglia**, segmentally arranged clusters of neurons.

Within an animal group, nervous system organization often correlates with lifestyle. Among the molluscs, for example, sessile and slow-moving species, such as clams and chitons, have relatively simple sense organs and little or no cephalization (**Figure 49.2f**). In contrast, active predatory molluscs, such as octopuses and squids (**Figure 49.2g**), have the most sophisticated nervous systems of any invertebrates, rivaling those of some vertebrates. With their large, image-forming eyes and a brain containing millions of neurons, octopuses can learn to discriminate between visual patterns and to perform complex tasks.

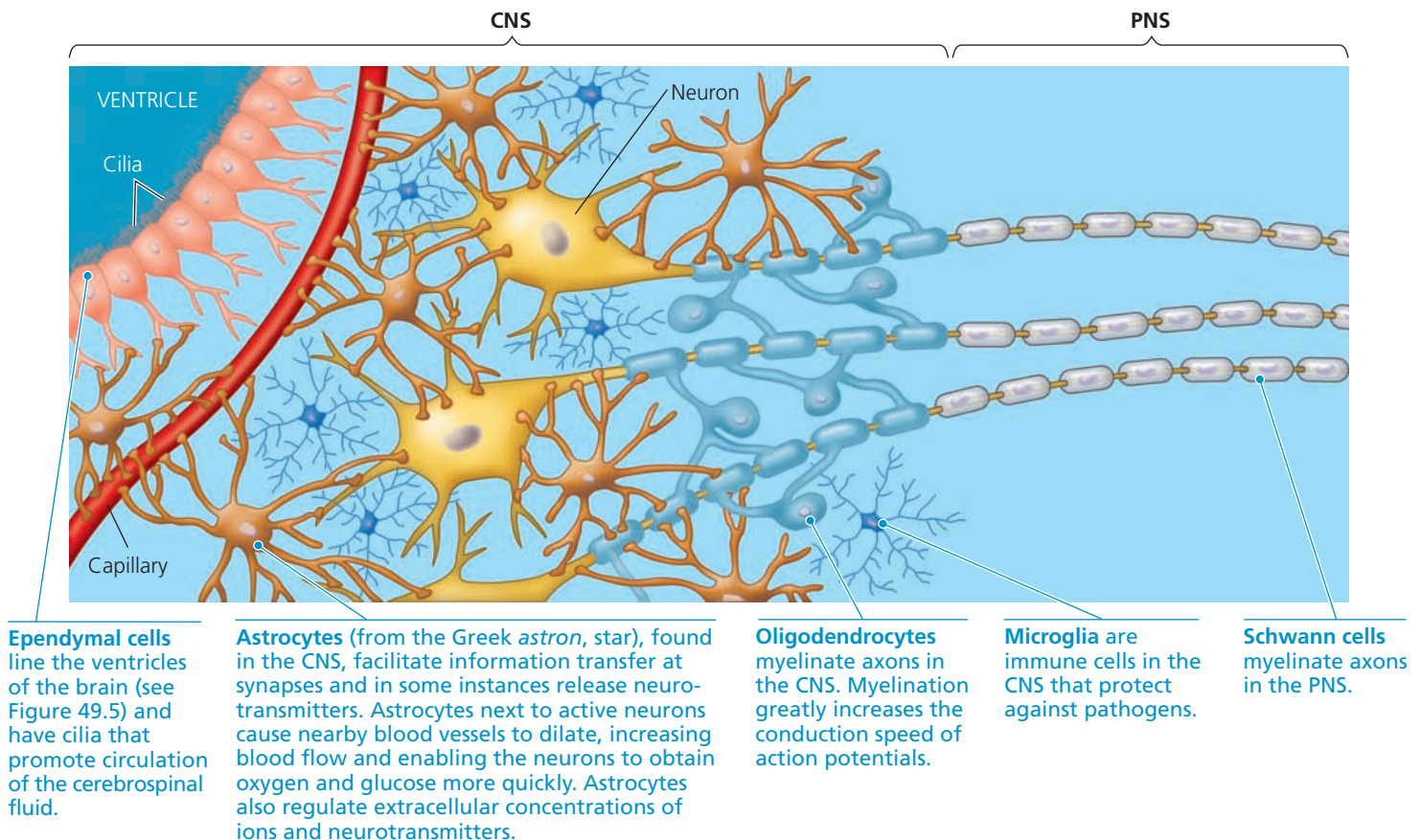
In vertebrates (**Figure 49.2h**), the brain and the spinal cord form the CNS; nerves and ganglia are the key components of the PNS. Regional specialization is a hallmark of both systems, as we will see throughout this chapter.

Glia

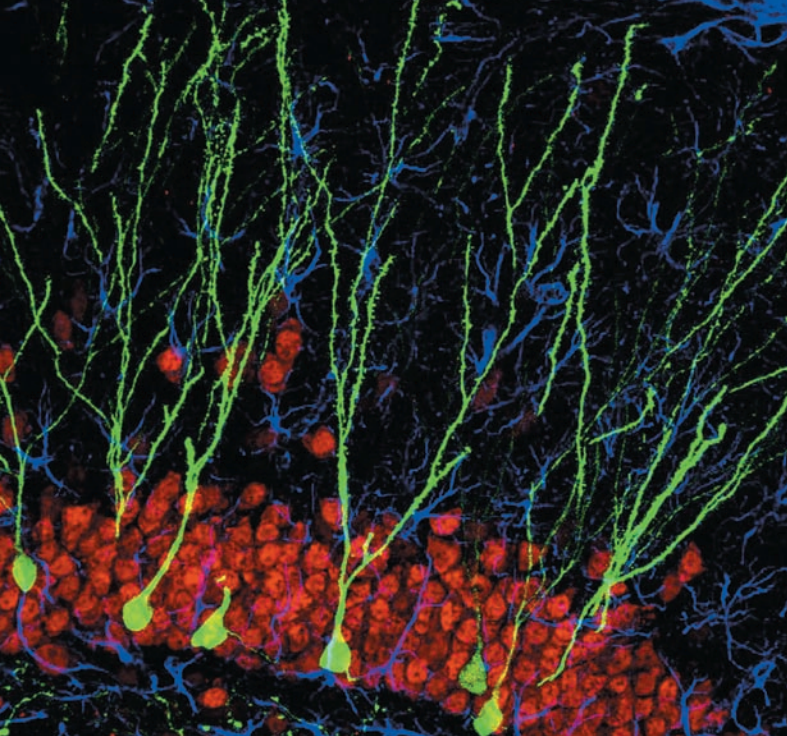
As discussed in Chapter 48, the nervous systems of vertebrates and most invertebrates include not only neurons but also **glial cells**, or **glia**. Some examples of glia are the Schwann cells that produce the myelin sheaths surrounding axons in the PNS and oligodendrocytes, their counterparts in the CNS. **Figure 49.3** illustrates the major types of glia in the adult vertebrate and provides an overview of the ways in which they nourish, support, and regulate the functioning of neurons.

One of the essential roles of glia is in nervous system development. In embryos, cells called *radial glia* form tracks along which newly formed neurons migrate from the neural tube, the structure that gives rise to the CNS (see Figure 47.14). Later, glia called **astrocytes** participate in formation of the *blood-brain barrier*, a specialization of the walls of brain capillaries that restricts the entry of most substances from the blood into the CNS.

Both radial glia and astrocytes can act as stem cells, which retain the ability to divide indefinitely. While some of their progeny remain undifferentiated, others differentiate into specialized cells. Studies with mice reveal that stem cells in the brain give rise to neurons that mature, migrate to particular locations, and become incorporated into the



▲ **Figure 49.3** Glia in the vertebrate nervous system.

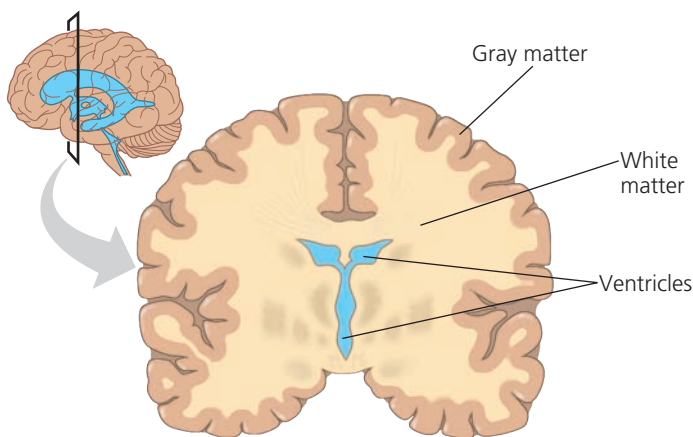


▲ **Figure 49.4** Newly born neurons in the brain of an adult mouse. In this light micrograph, new neurons derived from adult stem cells are labeled with green fluorescent protein (GFP), and all neurons are labeled with a DNA-binding dye, colored red in this image.

circuitry of the adult nervous system (**Figure 49.4**). Researchers are now tackling the challenge of finding a way to use neural stem cells as a means of replacing brain tissue that has ceased to function properly.

Organization of the Vertebrate Nervous System

During embryonic development in vertebrates, the central nervous system develops from the hollow dorsal nerve cord—a hallmark of chordates (see **Figure 34.3**). The cavity of the nerve cord gives rise to the narrow **central canal** of the spinal cord as well as the **ventricles** of the brain (**Figure 49.5**). Both



▲ **Figure 49.5** Ventricles, gray matter, and white matter. Ventricles deep in the brain's interior contain cerebrospinal fluid. Most of the gray matter is on the brain surface, surrounding the white matter.

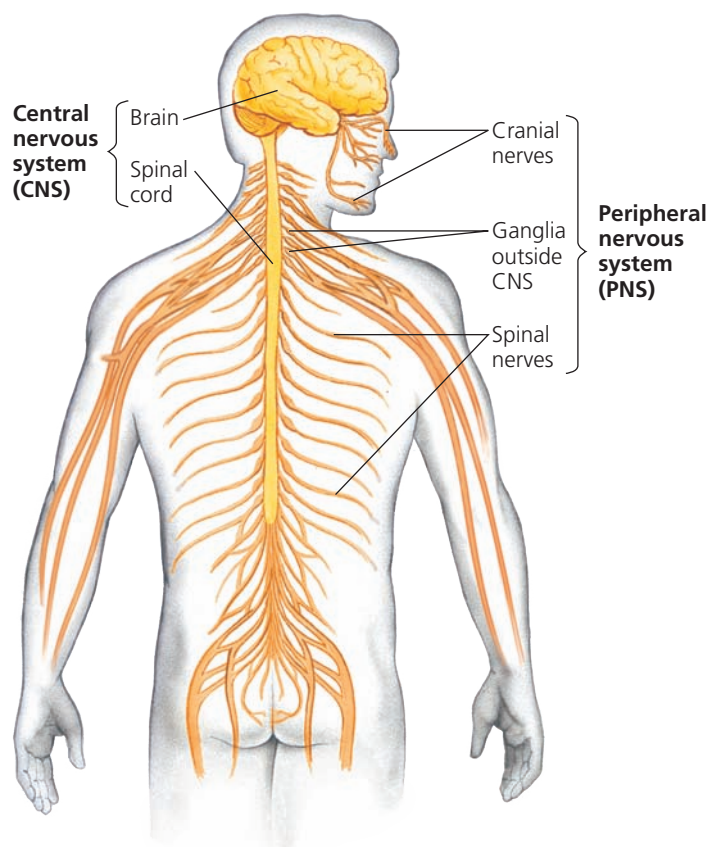
the canal and ventricles fill with *cerebrospinal fluid*, which is formed in the brain by filtration of arterial blood. The cerebrospinal fluid circulates slowly through the ventricles and central canal and then drains into the veins, supplying the CNS with nutrients and hormones and carrying away wastes.

In addition to these fluid-filled spaces, the brain and spinal cord contain gray matter and white matter (see **Figure 49.5**).

Gray matter is primarily made up of neuron cell bodies.

White matter consists mainly of bundled axons. In the spinal cord, white matter makes up the outer layer, consistent with its function in linking the CNS to sensory and motor neurons of the PNS. In the brain, white matter is predominantly in the interior, where signaling between neurons functions in learning, feeling emotions, processing sensory information, and generating commands.

In vertebrates, the spinal cord runs lengthwise inside the vertebral column, known as the spine (**Figure 49.6**). The spinal cord conveys information to and from the brain and generates basic patterns of locomotion. It also acts independently of the brain as part of the simple nerve circuits that produce **reflexes**, the body's automatic responses to certain stimuli.

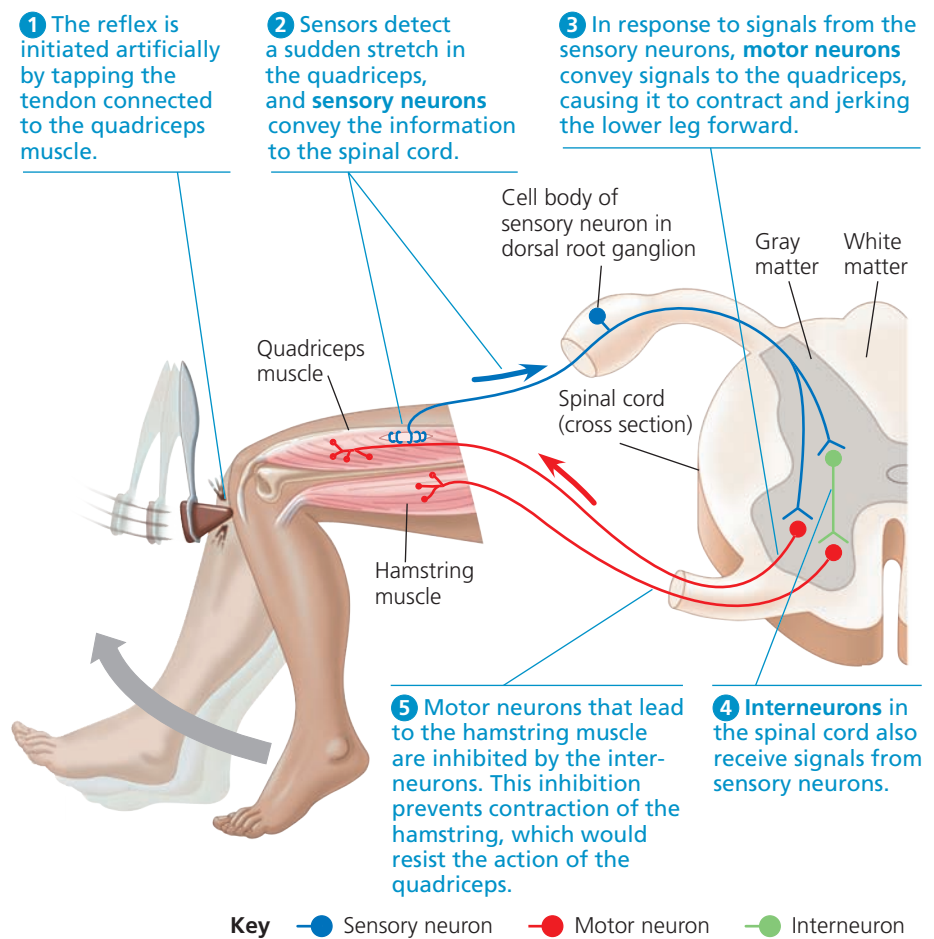


▲ **Figure 49.6** The vertebrate nervous system. The central nervous system consists of the brain and spinal cord (yellow). Left-right pairs of cranial nerves, spinal nerves, and ganglia make up most of the peripheral nervous system (dark gold).

► **Figure 49.7 The knee-jerk reflex.** Many neurons are involved in this reflex, but for simplicity, only a few neurons are shown.

MAKE CONNECTIONS Using the nerve signals to the hamstring and quadriceps in this reflex as an example, propose a model for regulation of smooth muscle activity in the esophagus during the swallowing reflex (see Figure 41.10).

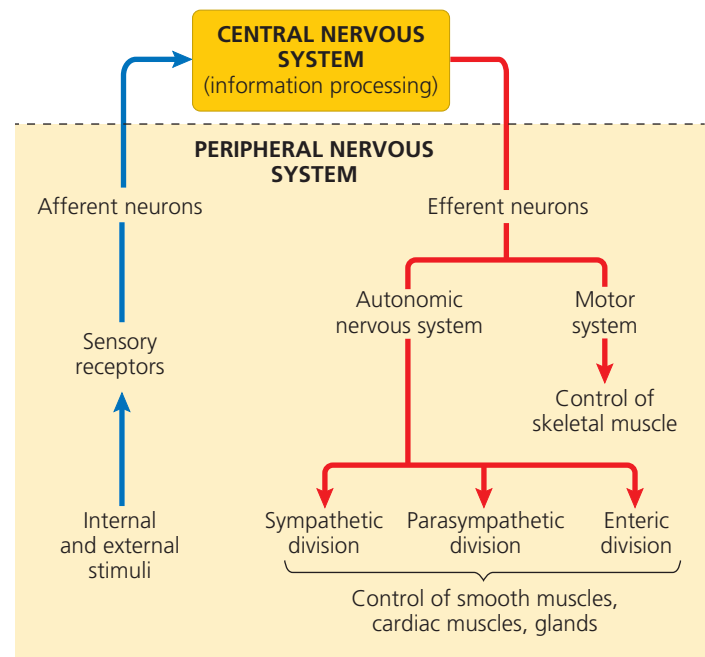
A reflex protects the body by providing a rapid, involuntary response to a particular stimulus. For example, if you accidentally put your hand on a hot burner, your hand begins to jerk back before your brain processes pain. Similarly, the knee-jerk reflex protects you when you pick up an unexpectedly heavy object. If your legs buckle, the tension across your knees triggers contraction of your thigh muscle (quadriceps), helping you stay upright and support the load. During a physical exam, your doctor may trigger the knee-jerk reflex with a triangular mallet to help assess nervous system function (**Figure 49.7**).



The Peripheral Nervous System

The PNS transmits information to and from the CNS and plays a large role in regulating both an animal's movement and its internal environment (**Figure 49.8**). Sensory information reaches the CNS along PNS neurons designated as *afferent* (from the Latin, meaning "to carry toward"). Following information processing within the CNS, instructions then travel to muscles, glands, and endocrine cells along PNS neurons designated as *efferent* (from the Latin, meaning "to carry away"). Most nerves contain both afferent and efferent neurons.

The PNS has two efferent components: the motor system and the autonomic nervous system (see Figure 49.8). The **motor system** consists of neurons that carry signals to skeletal muscles. Motor control can be voluntary, as when you raise your hand to ask a question, or involuntary, as in the knee-jerk reflex controlled by the spinal cord. In contrast, regulation of smooth and cardiac muscles by the **autonomic nervous system** is generally involuntary. The three divisions of the autonomic nervous system—sympathetic, parasympathetic, and enteric—together control the organs of the digestive, cardiovascular, excretory, and endocrine systems. For example, networks of neurons that form the **enteric division** of the autonomic nervous system are active in the digestive tract, pancreas, and gallbladder.



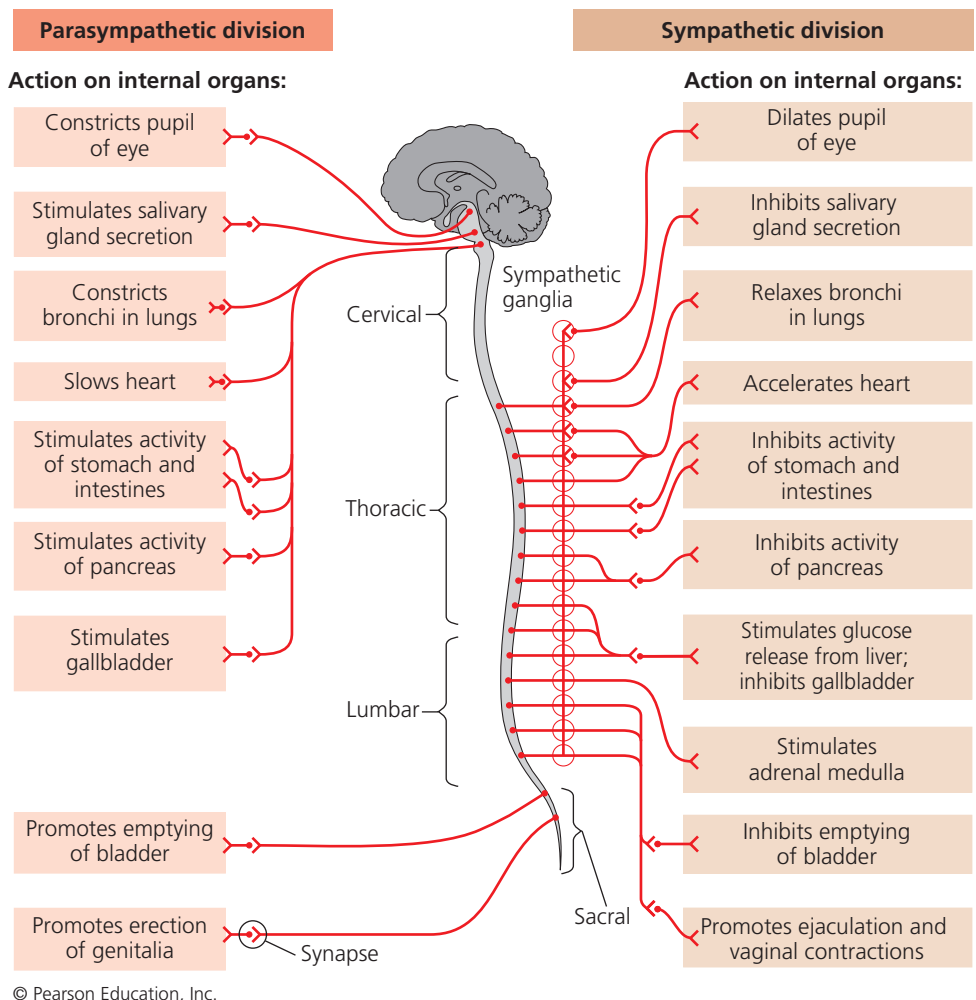
▲ **Figure 49.8 Functional hierarchy of the vertebrate peripheral nervous system.**

The sympathetic and parasympathetic divisions of the autonomic nervous system have largely antagonistic (opposite) functions in regulating organ function (**Figure 49.9**). Activation of the **sympathetic division** corresponds to arousal and energy generation (the “fight-or-flight” response). For example, the heart beats faster, digestion is inhibited, the liver converts glycogen to glucose, and the adrenal medulla increases secretion of epinephrine (adrenaline). Activation of the **parasympathetic division** generally causes opposite responses that promote calming and a return to self-maintenance functions (“rest and digest”). Thus, heart rate decreases, digestion is enhanced, and glycogen production increases. However, in regulating reproductive activity, a function that is not homeostatic, the parasympathetic division complements rather than antagonizes the sympathetic division (see **Figure 49.9**).

The two divisions differ not only in overall function but also in organization and signals released. Parasympathetic nerves exit the CNS at the base of the brain or spinal cord and form synapses in ganglia near or within an internal organ (see **Figure 49.9**). In contrast, sympathetic nerves typically exit the CNS midway along the spinal cord and form synapses in ganglia located just outside of the spinal cord.

In both the sympathetic and parasympathetic divisions, the pathway for information flow frequently involves a preganglionic and a postganglionic neuron. The *preganglionic neurons*, those with cell bodies in the CNS, release acetylcholine as a neurotransmitter. In the case of the *postganglionic neurons*, those of the parasympathetic division release acetylcholine, whereas their counterparts in the sympathetic division release norepinephrine. It is this difference in neurotransmitters that enables the sympathetic and parasympathetic divisions to bring about opposite effects in organs such as the lungs, heart, intestines, and bladder.

Homeostasis often relies on cooperation between the motor and autonomic nervous systems. In response to a drop in body temperature, for example, the hypothalamus signals the motor system to cause shivering, which increases heat production. At the same time, the hypothalamus signals the autonomic nervous system to constrict surface blood vessels, reducing heat loss.



▲ Figure 49.9 The parasympathetic and sympathetic divisions of the autonomic nervous system. Most pathways in each division involve two neurons. The axon of the first neuron extends from a cell body in the CNS to a set of PNS neurons whose cell bodies are clustered into a ganglion (plural, *ganglia*). The axons of these PNS neurons transmit instructions to internal organs, where they form synapses with smooth muscle, cardiac muscle, or gland cells.

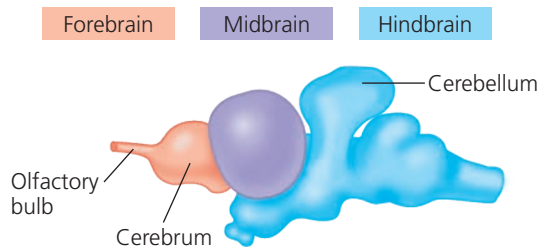
CONCEPT CHECK 49.1

- Which division of the autonomic nervous system would likely be activated if a student learned that an exam she had forgotten about would start in 5 minutes? Explain your answer.
- WHAT IF?** Suppose a person had an accident that severed a small nerve required to move some of the fingers of the right hand. Would you also expect an effect on sensation from those fingers?
- MAKE CONNECTIONS** Most tissues regulated by the autonomic nervous system receive both sympathetic and parasympathetic input from postganglionic neurons. Responses are typically local. In contrast, the adrenal medulla receives input only from the sympathetic division and only from preganglionic neurons, yet responses are observed throughout the body. Explain why (see **Figure 45.20**).

For suggested answers, see Appendix A.

The vertebrate brain is regionally specialized

We turn now to the vertebrate brain, which has three major regions: the forebrain, midbrain, and hindbrain (shown here for a ray-finned fish).



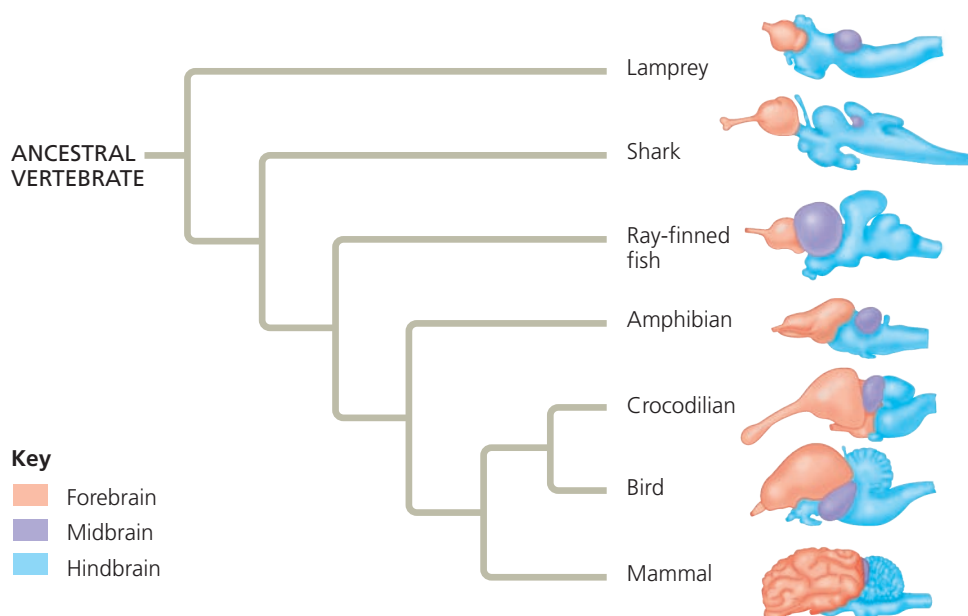
Each region is specialized in function. The **forebrain**, which contains the *olfactory bulb* and *cerebrum*, has activities that include processing of olfactory input (smells), regulation of sleep, learning, and any complex processing. The **midbrain**, located centrally in the brain, coordinates routing of sensory input. The **hindbrain**, part of which forms the *cerebellum*, controls involuntary activities, such as blood circulation, and coordinates motor activities, such as locomotion.

EVOLUTION Comparing vertebrates across a phylogenetic tree, we see that the relative sizes of particular brain regions vary (Figure 49.10). Furthermore, these size differences reflect differences in the importance of particular brain functions. Consider, for example, ray-finned fishes, which explore their environment using olfaction, vision,

and a lateral line system that detects water currents, electrical stimuli, and body position. The olfactory bulb, which detects scents in the water, is relatively large in these fishes. So is the midbrain, which processes input from the visual and lateral line systems. In contrast, the cerebrum, required for complex processing and learning, is relatively small. Evolution has thus resulted in a close match of structure to function, with the size of particular brain regions correlating with their importance for that species in nervous system function and, hence, species survival and reproduction.

The correlation between the size and function of brain regions can also be observed by considering the cerebellum. Free swimming ray-finned fishes, such as the tuna, control movement in three dimensions in the open water and have a relatively large cerebellum. In comparison, the cerebellum is much smaller in species that don't swim actively, such as the lamprey.

If one compares birds and mammals with groups that diverged from the common vertebrate ancestor earlier in evolution, two trends are apparent. First, the forebrain of birds and mammals occupies a larger fraction of the brain than it does in amphibians, fishes, and other vertebrates. Second, birds and mammals have much larger brains relative to body size than do other groups. Indeed, the ratio of brain size to body weight is ten times as large for birds and mammals as for their evolutionary ancestors. These differences in both overall brain size and the relative size of the forebrain reflect the greater capacity of birds and mammals for cognition and higher-order reasoning, traits we will return to later in this chapter.



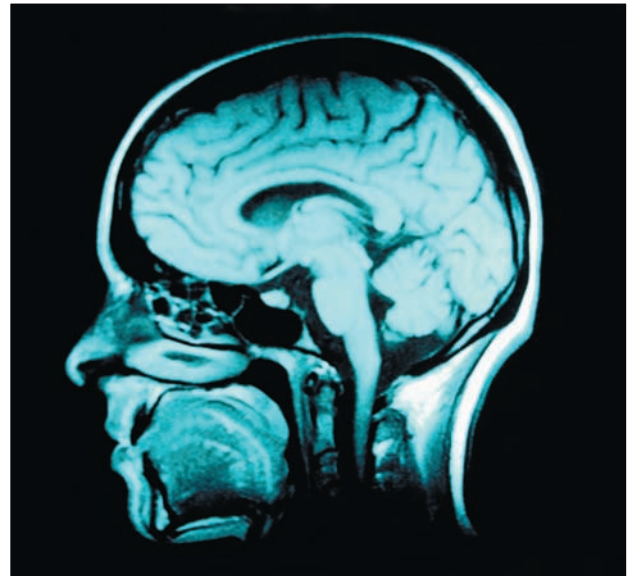
▲ Figure 49.10 Vertebrate brain structure and evolution. During evolution, differences arose in the relative size of the major structures common to vertebrate brains. As discussed in the text, size differences correlate with the importance of particular brain functions for particular vertebrate groups.

In the case of humans, it is estimated that the brain contains 100 billion neurons. How are so many cells organized into circuits and networks that can perform highly sophisticated information processing, storage, and retrieval? In addressing this question, let's begin with Figure 49.11, which explores the overall architecture of the human brain. You can use this figure to trace how brain structures arise during embryonic development; as a reference for their size, shape, and location in the adult brain; and as an introduction to their best-understood functions.

To learn more about how particular brain structure and brain organization overall relate to brain function in humans, we'll first consider activity cycles of the brain and the physiological basis of emotion. Then, in Concept 49.3, we'll shift our attention to regional specialization within the cerebrum.

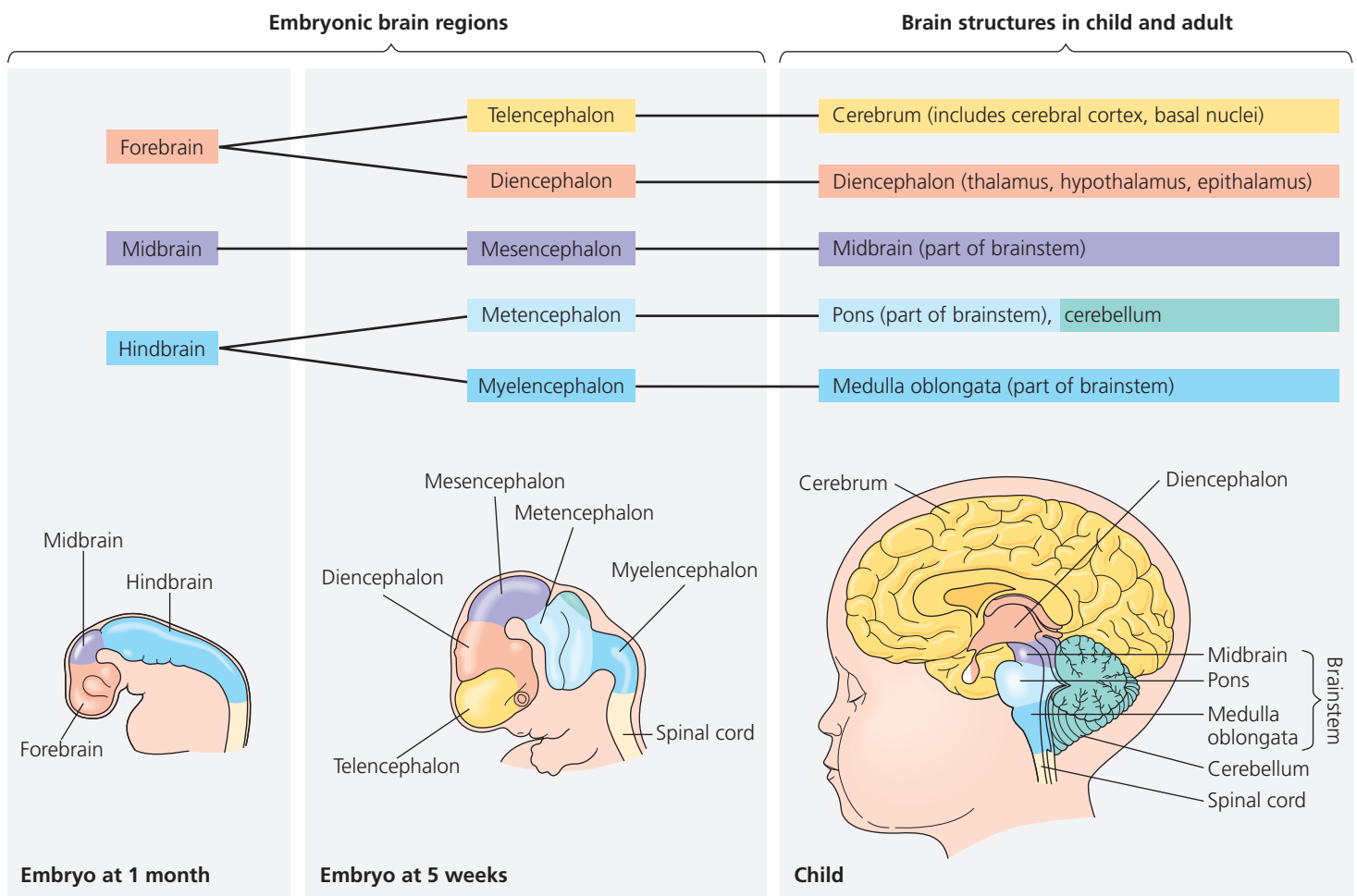
Exploring The Organization of the Human Brain

The brain is the most complex organ in the human body. Surrounded by the thick bones of the skull, the brain is divided into a set of distinctive structures, some of which are visible in the magnetic resonance image (MRI) of an adult's head shown at right. The diagram below traces the development of these structures in the embryo. Their major functions are explained on the facing page.



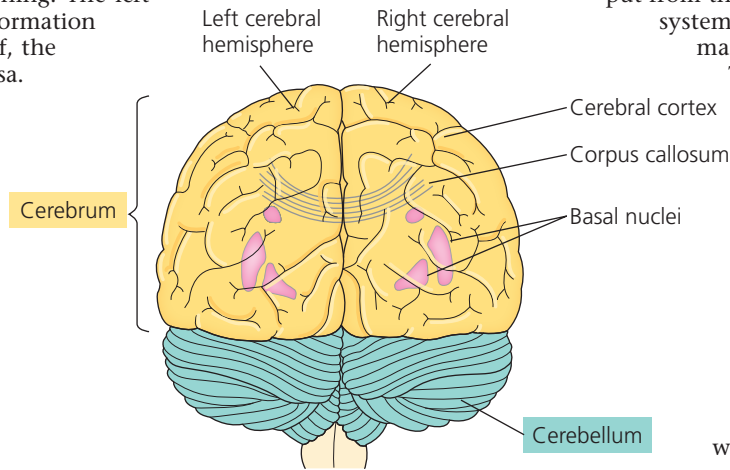
Human Brain Development

As a human embryo develops, the neural tube forms three anterior bulges—the forebrain, midbrain, and hindbrain—that together produce the adult brain. The midbrain and portions of the hindbrain give rise to the **brainstem**, a stalk that joins with the spinal cord at the base of the brain. The rest of the hindbrain gives rise to the **cerebellum**, which lies behind the brainstem. The third anterior bulge, the forebrain, develops into the diencephalon, including the neuroendocrine tissues of the brain, and the telencephalon, which becomes the **cerebrum**. Rapid, expansive growth of the telencephalon during the second and third months causes the outer portion, or cortex, of the cerebrum to extend over and around much of the rest of the brain.



The Cerebrum

The cerebrum controls skeletal muscle contraction and is the center for learning, emotion, memory, and perception. It is divided into right and left **cerebral hemispheres**. The outer layer of the cerebrum is called the **cerebral cortex** and is vital for perception, voluntary movement, and learning. The left side of the cerebral cortex receives information from, and controls the movement of, the right side of the body, and vice versa. A thick band of axons known as the **corpus callosum** enables the right and left cerebral cortices to communicate. Deep within the white matter, clusters of neurons called *basal nuclei* serve as centers for planning and learning movement sequences. Damage to these sites during fetal development can result in cerebral palsy, a disorder resulting from a disruption in the transmission of motor commands to the muscles.



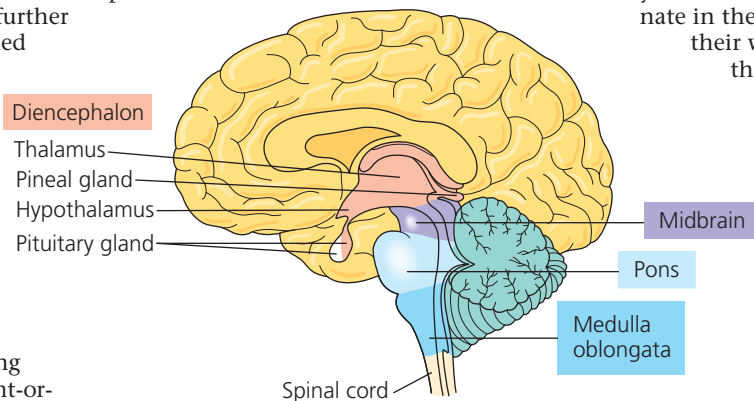
Adult brain viewed from the rear

The Cerebellum

The cerebellum coordinates movement and balance and helps in learning and remembering motor skills. The cerebellum receives sensory information about the positions of the joints and the lengths of the muscles, as well as input from the auditory (hearing) and visual systems. It also monitors motor commands issued by the cerebrum. The cerebellum integrates this information as it carries out coordination and error checking during motor and perceptual functions. Hand-eye coordination is an example of cerebellar control; if the cerebellum is damaged, the eyes can follow a moving object, but they will not stop at the same place as the object. Hand movement toward the object will also be erratic.

The Diencephalon

The diencephalon gives rise to the thalamus, hypothalamus, and epithalamus. The **thalamus** is the main input center for sensory information going to the cerebrum. Incoming information from all the senses, as well as from the cerebral cortex, is sorted in the thalamus and sent to the appropriate cerebral centers for further processing. The thalamus is formed by two masses, each roughly the size and shape of a walnut. A much smaller structure, the **hypothalamus**, constitutes a control center that includes the body's thermostat as well as the central biological clock. Through its regulation of the pituitary gland, the hypothalamus regulates hunger and thirst, plays a role in sexual and mating behaviors, and initiates the fight-or-flight response. The hypothalamus is also the source of posterior pituitary hormones and of releasing hormones that act on the anterior pituitary (see Figures 45.14 and 45.16). The *epithalamus* includes the pineal gland, the source of melatonin. It also contains one of several clusters of capillaries that generate cerebrospinal fluid from blood.



The Brainstem

The brainstem consists of the midbrain, the **pons**, and the **medulla oblongata** (commonly called the *medulla*). The midbrain receives and integrates several types of sensory information and sends it to specific regions of the forebrain. All sensory axons involved in hearing either terminate in the midbrain or pass through it on their way to the cerebrum. In addition, the midbrain coordinates visual reflexes, such as the peripheral vision reflex: The head turns toward an object approaching from the side without the brain having formed an image of the object. A major function of the pons and medulla is to transfer information between the PNS and the midbrain and forebrain. The pons and medulla also help coordinate large-scale body movements, such as running and climbing. Most axons that carry instructions about these movements cross from one side of the CNS to the other in the medulla. As a result, the right side of the brain controls much of the movement of the left side of the body, and vice versa. An additional function of the medulla is the control of several automatic, homeostatic functions, including breathing, heart and blood vessel activity, swallowing, vomiting, and digestion. The pons also participates in some of these activities; for example, it regulates the breathing centers in the medulla.

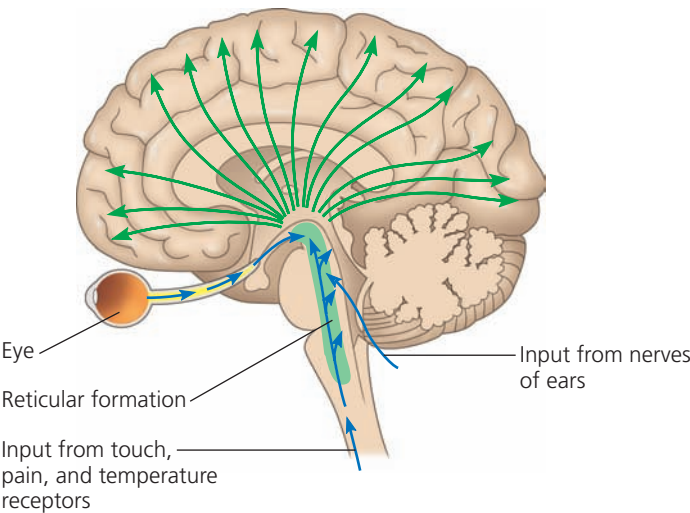
Arousal and Sleep

If you’ve ever drifted off to sleep while listening to a lecture (or reading a book), you know that your attentiveness and mental alertness can change rapidly. Such transitions are regulated by the brainstem and cerebrum, which control arousal and sleep. Arousal is a state of awareness of the external world. Sleep is a state in which external stimuli are received but not consciously perceived.

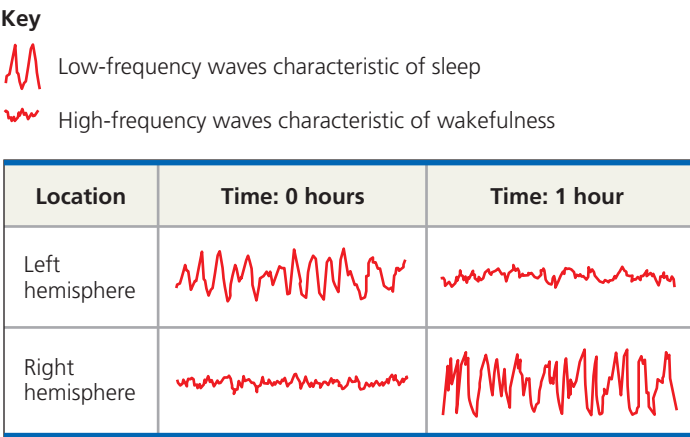
Contrary to appearances, sleep is an active state, at least for the brain. By placing electrodes at multiple sites on the scalp, we can record patterns of electrical activity called brain waves in an electroencephalogram (EEG). These recordings reveal that brain wave frequencies change as the brain progresses through distinct stages of sleep.

Although sleep is essential for survival, we still know very little about its function. One hypothesis is that sleep and dreams are involved in consolidating learning and memory. Evidence supporting this hypothesis includes the finding that test subjects who are kept awake for 36 hours have a reduced ability to remember when particular events occurred, even if they first “perk up” with caffeine. Other experiments show that regions of the brain that are activated during a learning task can become active again during sleep.

Arousal and sleep are controlled in part by the *reticular formation*, a diffuse network formed primarily by neurons in the midbrain and pons (Figure 49.12). These neurons control the timing of sleep periods characterized by rapid eye movements (REMs) and by vivid dreams. Sleep is also regulated by the biological clock and by regions of the forebrain that regulate sleep intensity and duration.



▲ **Figure 49.12 The reticular formation.** Once thought to consist of a single diffuse network of neurons, the reticular formation is now recognized as many distinct clusters of neurons. These clusters function in part to filter sensory input (blue arrows), blocking familiar and repetitive information that constantly enters the nervous system before sending the filtered input to the cerebral cortex (green arrows).



▲ **Figure 49.13 Dolphins can be asleep and awake at the same time.** EEG recordings were made separately for the two sides of a dolphin’s brain. At each time point, low-frequency activity was recorded in one hemisphere while higher-frequency activity typical of being awake was recorded in the other hemisphere.

Some animals have evolutionary adaptations that allow for substantial activity during sleep. Bottlenose dolphins, for example, swim while sleeping, rising to the surface to breathe air on a regular basis. How do they manage this feat? As in other mammals, the forebrain is physically and functionally divided into two halves, the right and left hemispheres. Noting that dolphins sleep with one eye open and one closed, researchers hypothesized that only one side of the brain is asleep at a time. EEG recordings from each hemisphere of sleeping dolphins support this hypothesis (Figure 49.13).

Biological Clock Regulation

Cycles of sleep and wakefulness are an example of a circadian rhythm, a daily cycle of biological activity. Such cycles, which occur in organisms ranging from bacteria to humans, rely on a **biological clock**, a molecular mechanism that directs periodic gene expression and cellular activity. Although biological clocks are typically synchronized to the cycles of light and dark in the environment, they can maintain a roughly 24-hour cycle even in the absence of environmental cues (see Figure 40.9). For example, in a constant environment humans exhibit a sleep/wake cycle of 24.2 hours, with very little variation among individuals.

What normally links an animal’s biological clock to environmental cycles of light and dark? In mammals, circadian rhythms are coordinated by a group of neurons in the hypothalamus called the **suprachiasmatic nucleus**, or SCN. (Certain clusters of neurons in the CNS are referred to as “nuclei.”) In response to sensory information from the eyes, the SCN acts as a pacemaker, synchronizing the biological clock in cells throughout the body to the natural cycles of day length. In the **Scientific Skills Exercise**, you can interpret data from an experiment and propose experiments to test the role of the SCN in hamster circadian rhythms.

SCIENTIFIC SKILLS EXERCISE

Designing an Experiment Using Genetic Mutants

Does the SCN Control the Circadian Rhythm in Hamsters?

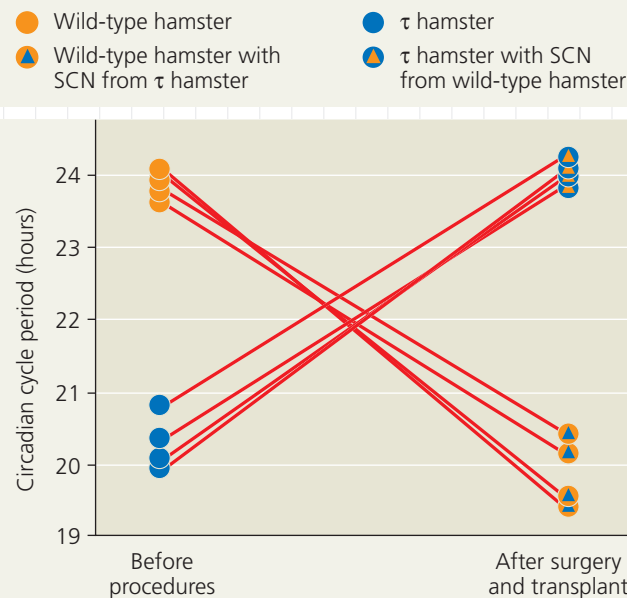
By surgically removing the SCN from laboratory mammals, scientists demonstrated that the SCN is required for circadian rhythms. Those experiments did not, however, reveal whether circadian rhythms originate in the SCN. To answer this question, researchers performed an SCN transplant experiment on wild-type and mutant hamsters (*Mesocricetus auratus*). Whereas wild-type hamsters have a circadian cycle lasting about 24 hours in the absence of external cues, hamsters homozygous for the τ (tau) mutation have a cycle lasting only about 20 hours. In this exercise, you will evaluate the design of this experiment and propose additional experiments to gain further insight.

How the Experiment Was Done The researchers surgically removed the SCN from wild-type and τ hamsters. Several weeks later, each of these hamsters received a transplant of an SCN from a hamster of the opposite genotype. To determine the periodicity of rhythmic activity for the hamsters before the surgery and after the transplants, the researchers measured activity levels over a three-week period. They plotted the data collected for each day in the manner shown in Figure 40.9a and then calculated the circadian cycle period.

Data from the Experiment In 80% of the hamsters in which the SCN had been removed, transplanting an SCN from another hamster restored rhythmic activity. For hamsters in which an SCN transplant restored a circadian rhythm, the net effect of the two procedures (SCN removal and replacement) on the circadian cycle period is graphed at the upper right. Each red line connects the two data points for an individual hamster.

Interpret the Data

1. In a controlled experiment, researchers manipulate one variable at a time. What was the variable manipulated in this study? Why did the researchers use more than one hamster for each procedure? What traits of the individual hamsters would likely have been held constant among the treatment groups?
2. For the wild-type hamsters that received τ SCN transplants, what would have been an appropriate experimental control?



3. What general trends does the graph above reveal about the circadian cycle period of the transplant recipients? Do the trends differ for the wild-type and τ recipients? Based on these data, what can you conclude about the role of the SCN in determining the period of the circadian rhythm?
4. In 20% of the hamsters, there was no restoration of rhythmic activity following the SCN transplant. What are some possible reasons for this finding? Do you think you can be confident of your conclusion about the role of the SCN based on data from 80% of the hamsters?
5. Suppose that researchers identified a mutant hamster that lacked rhythmic activity; that is, its circadian activity cycle had no regular pattern. Propose SCN transplant experiments using such a mutant along with (a) wild-type and (b) τ hamsters. Predict the results of those experiments in light of your conclusion in question 3.

A version of this Scientific Skills Exercise can be assigned in MasteringBiology.

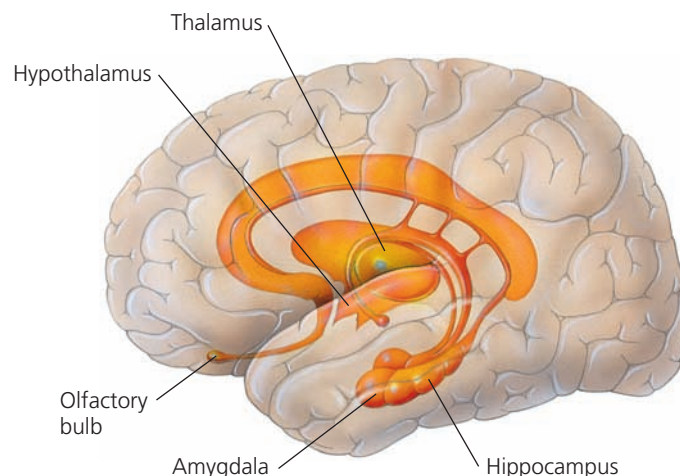
Data from M. R. Ralph et al., Transplanted suprachiasmatic nucleus determines circadian period, *Science* 247:975–978 (1990). Reprinted with permission from AAAS.

Emotions

Whereas a single structure in the brain controls the biological clock, the generation and experience of emotions depend on many brain structures, including the amygdala, hippocampus, and parts of the thalamus. As shown in Figure 49.14, these structures border the brainstem in mammals and are therefore called the *limbic system* (from the Latin *limbus*, border).

Generating emotion and experiencing emotion often require interactions between different regions of the brain. For example, laughing and crying both involve the limbic system interacting with sensory areas of the forebrain. Similarly, structures in the forebrain attach emotional “feelings” to survival-related functions controlled by the brainstem, including aggression, feeding, and sexuality.

Emotional experiences are often stored as memories that can be recalled by similar circumstances. For example, a situation that causes you to remember a frightening event



▲ Figure 49.14 The limbic system in the human brain.

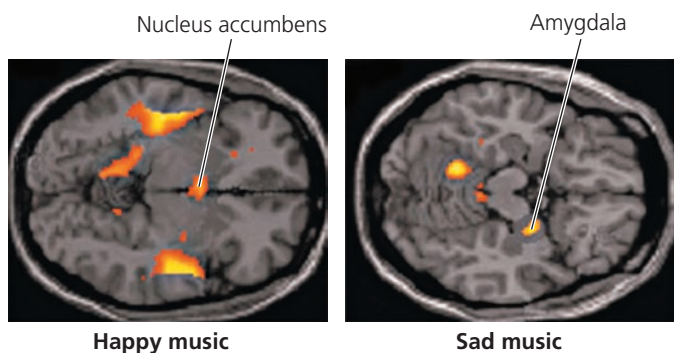
can trigger a faster heart rate, sweating, and mental state of fear, even if there is currently nothing scary or threatening in your surroundings. The brain structure that is most important for this emotional memory is the **amygdala**, an almond-shaped mass of nuclei (clusters of neurons) located near the base of the cerebrum.

To study the function of the human amygdala, researchers sometimes present adult subjects with an image followed by an unpleasant experience, such as a mild electrical shock. After several trials, study participants experience *autonomic arousal*—as measured by increased heart rate or sweating—if they see the image again. Subjects with brain damage confined to the amygdala can recall the image because their explicit memory is intact. However, they do not exhibit autonomic arousal, indicating that damage to the amygdala has resulted in a reduced capacity for emotional memory.

Functional Imaging of the Brain

Today, the amygdala and other brain structures are being probed and analyzed with functional imaging methods. The first widely used technique was positron-emission tomography (PET), in which injection of radioactive glucose enables a display of metabolic activity. Today, many studies rely on functional magnetic resonance imaging (fMRI). In fMRI, a subject lies with his or her head in the center of a large, doughnut-shaped magnet. Brain activity in a region is detected by changes in the local oxygen concentration. By scanning the brain while the subject performs a task, such as forming a mental image of a person's face, researchers can correlate particular tasks with activity in specific brain areas.

In one experiment using fMRI, researchers mapped brain activity while subjects listened to music that they described as happy or sad (**Figure 49.15**). Listening to happy music led to increased activity in the nucleus accumbens, a brain



▲ **Figure 49.15** Functional imaging in the working brain. Functional magnetic resonance imaging (fMRI) was used to reveal brain activity associated with music that listeners described as happy or sad.

WHAT IF? In the experiment that produced the images shown above, some regions of the brain were active under both conditions. What function might such regions carry out?

structure important for the perception of pleasure. In contrast, subjects who heard sad music had increased activity in the amygdala.

The range of applications of fMRI includes monitoring recovery from stroke, mapping abnormalities in migraine headaches, and increasing the effectiveness of brain surgery. This technique has even been used to explore sex-based differences in the CNS, demonstrating, for instance, that cerebral blood flow is higher on average in women than in men.

CONCEPT CHECK 49.2

1. When you wave your right hand, what part of your brain initiates the action?
2. People who are inebriated have difficulty touching their nose with their eyes closed. Which brain region does this observation indicate is one of those impaired by alcohol?
3. **WHAT IF?** Suppose you examine two groups of individuals with CNS damage. In one group, the damage has resulted in a coma (a prolonged state of unconsciousness). In the other group, it has caused paralysis (a loss of skeletal muscle function throughout the body). Relative to the position of the midbrain and pons, where is the likely site of damage in each group? Explain.

For suggested answers, see Appendix A.

CONCEPT 49.3

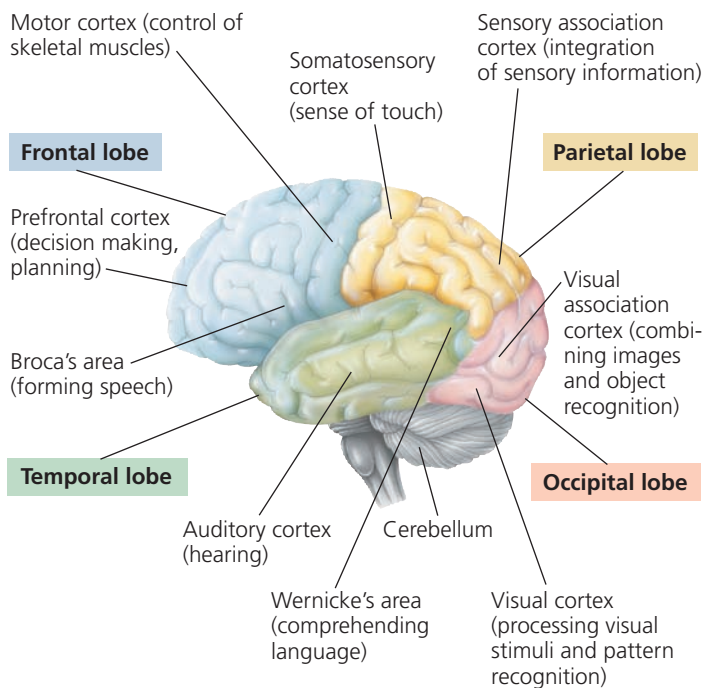
The cerebral cortex controls voluntary movement and cognitive functions

We turn now to the cerebrum, the part of the brain essential for language, cognition, memory, consciousness, and awareness of our surroundings. As shown in Figure 49.11, the cerebrum is the largest structure in the human brain. Like the brain overall, it exhibits regional specialization. For the most part, cognitive functions reside in the cortex, the outer layer of the cerebrum. Within the cortex, *sensory areas* receive and process sensory information, *association areas* integrate the information, and *motor areas* transmit instructions to other parts of the body.

In discussing the location of particular functions in the cerebral cortex, neurobiologists often use four regions, or *lobes*, as physical landmarks. As shown in **Figure 49.16**, each side of the cerebral cortex has a frontal, temporal, occipital, and parietal lobe (each is named for a nearby bone of the skull).

Information Processing

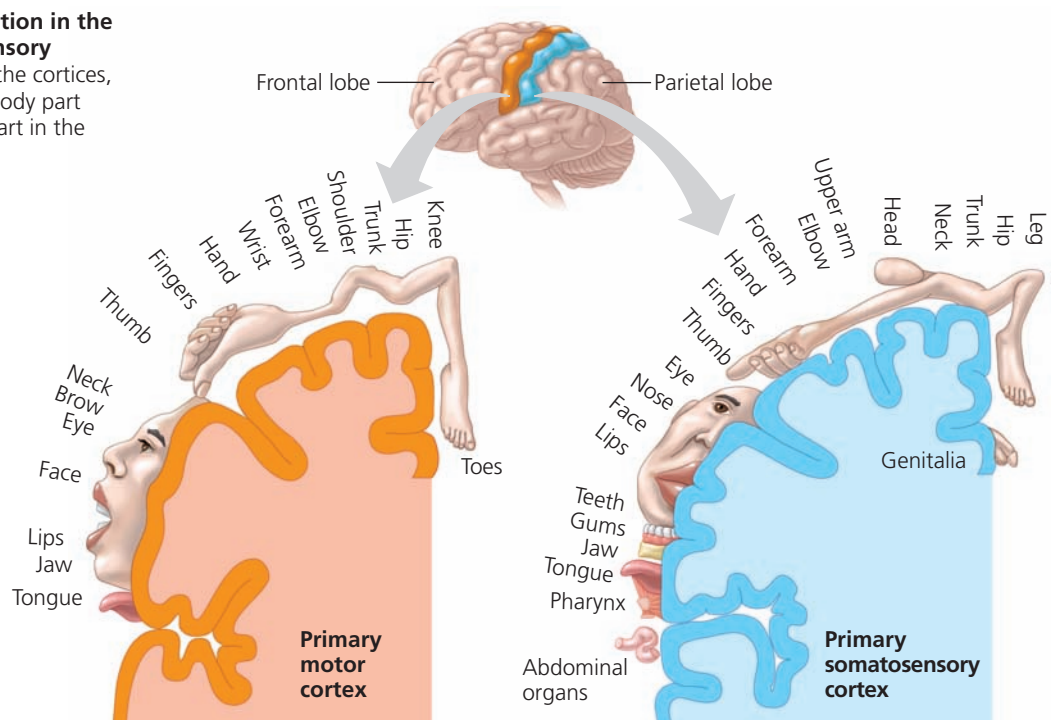
Broadly speaking, there are two sources of sensory input to the human cerebral cortex. Some sensory input comes from groups of receptors clustered in dedicated sensory organs, such as the eyes and nose. Other sensory input originates



▲ **Figure 49.16 The human cerebral cortex.** Each side of the cerebral cortex is divided into four lobes, and each lobe has specialized functions, some of which are listed here. Some areas on the left side of the brain (shown here) have different functions from those on the right side (not shown).

in individual receptors in the hands, scalp, and elsewhere in the body. These somatic sensory, or *somatosensory*, receptors (from the Greek *soma*, body) provide information about touch, pain, pressure, temperature, and the position of muscles and limbs.

► **Figure 49.17 Body part representation in the primary motor and primary somatosensory cortices.** In these cross-sectional maps of the cortices, the cortical surface area devoted to each body part is represented by the relative size of that part in the cartoons.



Most sensory information coming into the cortex is directed via the thalamus to primary sensory areas within the brain lobes. Information received at the primary sensory areas is passed along to nearby association areas, which process particular features in the sensory input. In the occipital lobe, for instance, some groups of neurons in the primary visual area are specifically sensitive to rays of light oriented in a particular direction. In the visual association area, information related to such features is combined in a region dedicated to recognizing complex images, such as faces.

Once processed, sensory information passes to the prefrontal cortex, which helps plan actions and movement. The cerebral cortex may then generate motor commands that cause particular behaviors—moving a limb or saying hello, for example. These commands consist of action potentials produced by neurons in the motor cortex, which lies at the rear of the frontal lobe (see Figure 49.16). The action potentials travel along axons to the brainstem and spinal cord, where they excite motor neurons, which in turn excite skeletal muscle cells.

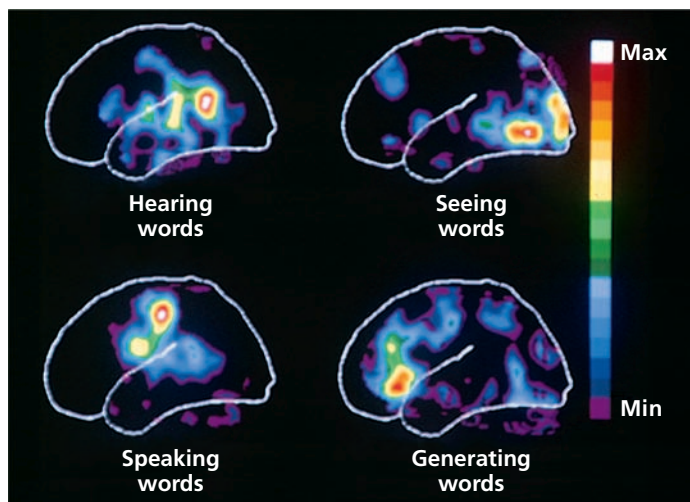
In the somatosensory cortex and motor cortex, neurons are arranged according to the part of the body that generates the sensory input or receives the motor commands (**Figure 49.17**). For example, neurons that process sensory information from the legs and feet lie in the region of the somatosensory cortex closest to the midline. Neurons that control muscles in the legs and feet are located in the corresponding region of the motor cortex. Notice in Figure 49.17 that the cortical surface area devoted to each body part is not proportional to the size of the part. Instead, surface area correlates with the extent of neuronal control needed

(for the motor cortex) or with the number of sensory neurons that extend axons to that part (for the somatosensory cortex). Thus, the surface area of the motor cortex devoted to the face is much larger than that devoted to the trunk, reflecting the extensive involvement of facial muscles in communication.

Although our focus here is on humans, it is worth noting that the processing sites for sensory information vary among vertebrates. In ray-finned fishes, for example, the relatively large midbrain (see Figure 49.10) serves as the primary center for processing and responding to visual stimuli. Such differences reflect a recognizable evolutionary trend: Following the vertebrate phylogenetic tree from sharks to ray-finned fishes, amphibians, reptiles, and finally mammals, one observes a steadily increasing role for the forebrain in processing sensory information.

Language and Speech

The mapping of cognitive functions within the cortex began in the 1800s when physicians studied the effects of damage to particular regions of the cortex by injuries, strokes, or tumors. Pierre Broca conducted postmortem (after death) examinations of patients who had been able to understand language but unable to speak. He discovered that many had defects in a small region of the left frontal lobe, now known as *Broca's area*. Karl Wernicke found that damage to a posterior portion of the left temporal lobe, now called *Wernicke's area*, abolished the ability to comprehend speech but not the ability to speak. PET studies have now confirmed activity in Broca's area during speech generation and Wernicke's area when speech is heard (**Figure 49.18**).



▲ **Figure 49.18** Mapping language areas in the cerebral cortex. These PET images show regions with different activity levels in one person's brain during four activities, all related to speech. Increases in activity are seen in Wernicke's area when hearing words, Broca's area when speaking words, the visual cortex when seeing words, and the frontal lobe when generating words (without reading them).

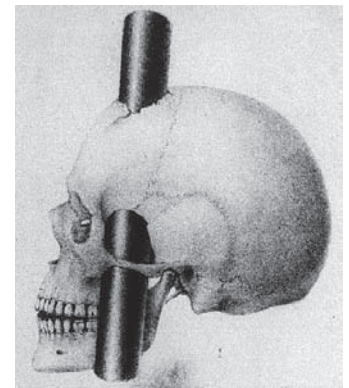
Lateralization of Cortical Function

Both Broca's area and Wernicke's area reside in the left cortical hemisphere, reflecting a greater role with regard to language for the left side of the cerebrum than for the right side. The left hemisphere is also more adept at math and logical operations. In contrast, the right hemisphere appears to be dominant in the recognition of faces and patterns, spatial relations, and nonverbal thinking. The establishment of these differences in hemisphere function is called **lateralization**.

The two cortical hemispheres normally exchange information through the fibers of the corpus callosum (see Figure 49.11). Severing this connection (a treatment of last resort for the most extreme forms of epilepsy, a seizure disorder) results in a "split-brain" effect. In such patients, the two hemispheres function independently. For example, they cannot read even a familiar word that appears only in their left field of vision: The sensory information that travels from the left field of vision to the right hemisphere cannot reach the language centers in the left hemisphere.

Frontal Lobe Function

In 1848, a horrific accident pointed to the role of the prefrontal cortex in temperament and decision making. Phineas Gage was the foreman of a railroad construction crew when an explosion drove an iron rod through his head. The rod, which was more than 3 cm in diameter at one end, entered his skull just below his left eye and exited through the top of his head, damaging large portions of his frontal lobe. Gage recovered, but his personality changed dramatically. He became emotionally detached, impatient, and erratic in his behavior.



Some frontal lobe tumors cause symptoms similar to those of Gage's brain injury. Intellect and memory seem intact, but decision making is flawed and emotional responses are diminished. In the 1900s, the same problems resulted from frontal lobotomy, a surgical procedure that severs the connection between the prefrontal cortex and the limbic system. Together, these observations provide evidence that the frontal lobes have a substantial influence on what are called "executive functions."

Once a common treatment for severe behavioral disorders, frontal lobotomy is no longer in use. Instead, behavioral disorders are typically treated with medications, as discussed later in this chapter.

Evolution of Cognition in Vertebrates

EVOLUTION In nearly all vertebrates, the brain has the same basic structures (see Figure 49.10). Given this uniform organization, how did a capacity for advanced cognition, the perception and reasoning that constitute knowledge, evolve in certain species? One hypothesis is that higher-order reasoning required evolution of an extensively convoluted cerebral cortex, as is found in humans, other primates, and cetaceans (whales, dolphins, and porpoises). Indeed, in humans the cerebral cortex accounts for about 80% of total brain mass.

Birds, on the other hand, lack a convoluted cerebral cortex and were therefore thought to have much lower intellectual capacity than primates and cetaceans. However, experiments in recent years have refuted this idea. Western scrub jays (*Aphelocoma californica*) can remember which food items they hid first. New Caledonian crows (*Corvus moneduloides*) are highly skilled at making and using tools, an ability otherwise well documented only for humans and some other apes. Furthermore, African gray parrots (*Psittacus erithacus*) understand numerical and abstract concepts, such as “same” and “different” and “none.”

The anatomical basis for sophisticated information processing in birds appears to be a clustered organization of neurons within the *pallium*, the top or outer portion of the brain (Figure 49.19a). This arrangement is different from

that in the human cerebral cortex, where six parallel layers of neurons are arranged tangential to the brain surface (Figure 49.19b). Thus, evolution has resulted in two types of outer brain organization in vertebrates that support complex and flexible brain function.

How did the bird pallium and human cerebral cortex arise during evolution? The current consensus is that the common ancestor of birds and mammals had a pallium in which neurons were organized into nuclei, as is still found in birds. Early in mammalian evolution, this clustered organization was transformed into a layered one. However, connectivity was maintained such that, for example, the thalamus relays sensory input relating to sights, sounds, and touch to the pallium in birds and the cerebral cortex in mammals.

Sophisticated information processing depends not only on the overall organization of a brain but also on the very small-scale changes that enable learning and encode memory. We'll turn to these changes in the context of humans in the next section.

CONCEPT CHECK 49.3

1. How can studying individuals with damage to a particular brain region provide insight into the normal function of that region?
2. How do the functions of Broca's area and Wernicke's area each relate to the activity of the surrounding cortex?
3. **WHAT IF?** If a woman with a severed corpus callosum viewed a photograph of a familiar face, first in her left field of vision and then in her right field, why would she find it difficult to put a name to the face?

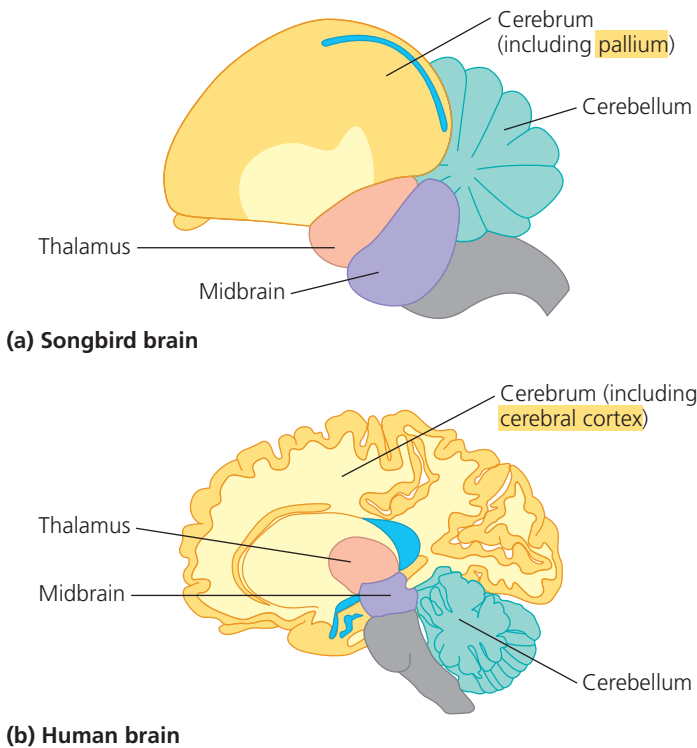
For suggested answers, see Appendix A.

CONCEPT 49.4

Changes in synaptic connections underlie memory and learning

During embryonic development, regulated gene expression and signal transduction establish the overall structure of the nervous system (see Chapter 47). Two processes then dominate the remaining development and remodeling of the nervous system. The first is a competition among neurons for survival. Neurons compete for growth-supporting factors, which are produced in limited quantities by tissues that direct neuron growth. Cells that don't reach the proper locations fail to receive such factors and undergo programmed cell death. The competition is so severe that half of the neurons formed in the embryo are eliminated. The net effect is the preferential survival of neurons that are located properly within the nervous system.

Synapse elimination is the second major process that shapes the nervous system. A developing neuron forms numerous synapses, more than are required for its proper



▲ **Figure 49.19** Comparison of regions for higher cognition in avian and human brains. Although structurally different, the pallium of a songbird brain (a) and the cerebral cortex of the human brain (b) play similar roles in higher cognitive activities and make many similar connections with other brain structures.

function. The activity of that neuron then stabilizes some synapses and destabilizes others. By the end of embryonic development, more than half of all synapses have been eliminated.

Together, neuron death and synapse elimination set up the basic network of cells and connections within the nervous system required throughout life.

Neuronal Plasticity

Although the overall organization of the CNS is established during embryonic development, the connections between neurons can be modified. This capacity for the nervous system to be remodeled, especially in response to its own activity, is called **neuronal plasticity**.

Much of the reshaping of the nervous system occurs at synapses. When the activity of a synapse coincides with that of other synapses, changes may occur that reinforce that synaptic connection. Conversely, when the activity of a synapse fails to correlate in this way with that of other synapses, the synaptic connection sometimes becomes weaker. In this way, synapses belonging to circuits that link information in useful ways are maintained, whereas those that convey bits of information lacking any context are lost.

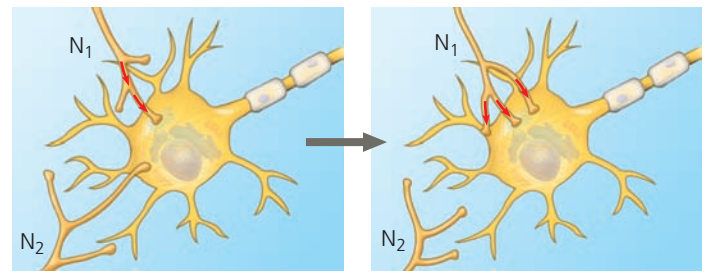
Figure 49.20a illustrates how activity-dependent events can result in either the addition or loss of a synapse. If you think of signals in the nervous system as traffic on a highway, such changes are comparable to adding or removing an entrance ramp. The net effect is to increase signaling between particular pairs of neurons and decrease signaling between other pairs. As shown in **Figure 49.20b**, changes can also strengthen or weaken signaling at a synapse. In our traffic analogy, this would be equivalent to widening or narrowing an entrance ramp.

Research indicates that *autism*, a developmental disorder that first appears early in childhood, involves a disruption of activity-dependent remodeling at synapses. Children affected with autism display impaired communication and social interaction, as well as stereotyped and repetitive behaviors.

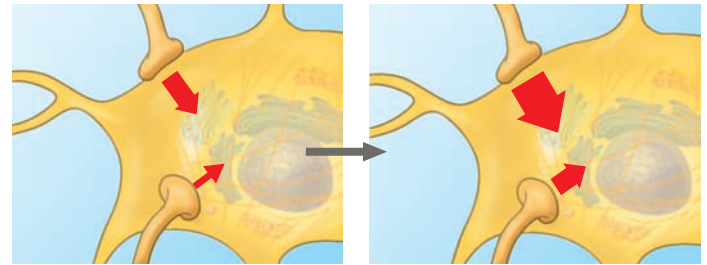
Although the underlying causes of autism are unknown, there is a strong genetic contribution to this and related disorders. Extensive research has ruled out a link to vaccine preservatives, once proposed as a potential risk factor. Further understanding of the autism-associated disruption in synaptic plasticity may help efforts to better understand and treat this disorder.

Memory and Learning

Neuronal plasticity is essential to the formation of memories. We are constantly checking what is happening against what just happened. We hold information for a time in **short-term**



(a) Connections between neurons are strengthened or weakened in response to activity. High-level activity at the synapse of the postsynaptic neuron with presynaptic neuron N_1 leads to recruitment of additional axon terminals from that neuron. Lack of activity at the synapse with presynaptic neuron N_2 leads to loss of functional connections with that neuron.



(b) If two synapses on the same postsynaptic cell are often active at the same time, the strength of the postsynaptic response may increase at both synapses.

▲ **Figure 49.20 Neuronal plasticity.** Synaptic connections can change over time, depending on the activity level at the synapse.

memory and then release it if it becomes irrelevant. If we wish to retain knowledge of a name, phone number, or other fact, the mechanisms of **long-term memory** are activated. If we later need to recall the name or number, we fetch it from long-term memory and return it to short-term memory.

Both short-term and long-term memory involve the storage of information in the cerebral cortex. In short-term memory, this information is accessed via temporary links formed in the hippocampus. When memories are made long-term, the links in the hippocampus are replaced by connections within the cerebral cortex itself. As discussed earlier, some of this consolidation of memory is thought to occur during sleep. Furthermore, the reactivation of the hippocampus that is required for memory consolidation likely forms the basis for at least some of our dreams.

According to our current understanding of memory, the hippocampus is essential for acquiring new long-term memories but not for maintaining them. This hypothesis readily explains the symptoms of some individuals who suffer damage to the hippocampus: They cannot form any new lasting memories but can freely recall events from before their injury. In effect, their lack of normal hippocampal function traps them in their past.

What evolutionary advantage might be offered by organizing short-term and long-term memories differently? One

hypothesis is that the delay in forming connections in the cerebral cortex allows long-term memories to be integrated gradually into the existing store of knowledge and experience, providing a basis for more meaningful associations. Consistent with this hypothesis, the transfer of information from short-term to long-term memory is enhanced by the association of new data with data previously learned and stored in long-term memory. For example, it's easier to learn a new card game if you already have "card sense" from playing other card games.

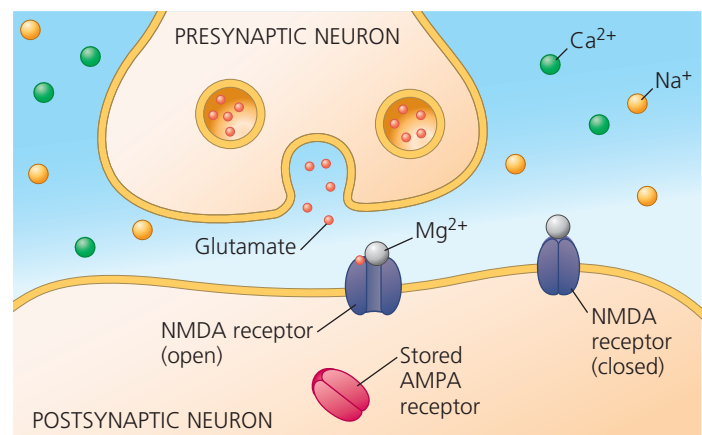
Motor skills, such as tying your shoes or writing, are usually learned by repetition. You can perform these skills without consciously recalling the individual steps required to do these tasks correctly. Learning skills and procedures, such as those required to ride a bicycle, appears to involve cellular mechanisms very similar to those responsible for brain growth and development. In such cases, neurons actually make new connections. In contrast, memorizing phone numbers, facts, and places—which can be very rapid and may require only one exposure to the relevant item—may rely mainly on changes in the strength of existing neuronal connections. Next we will consider one way that such changes in strength can take place.

Long-Term Potentiation

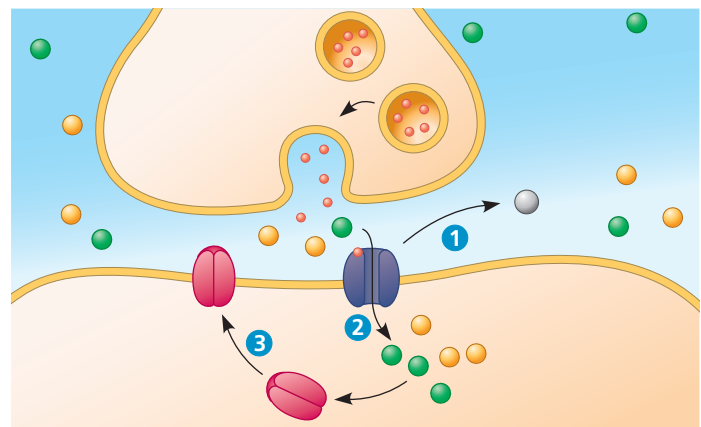
In searching for the physiological basis of memory, researchers have concentrated their attention on processes that can alter a synaptic connection, making the flow of communication either more efficient or less efficient. We will focus here on **long-term potentiation (LTP)**, a lasting increase in the strength of synaptic transmission.

First characterized in tissue slices from the hippocampus, LTP involves a presynaptic neuron that releases the excitatory neurotransmitter glutamate. For LTP to occur, there must be a high-frequency series of action potentials in this presynaptic neuron. In addition, these action potentials must arrive at the synaptic terminal at the same time that the postsynaptic cell receives a depolarizing stimulus at another synapse. The net effect is to strengthen a synapse whose activity coincides with that of another input (see Figure 48.17a).

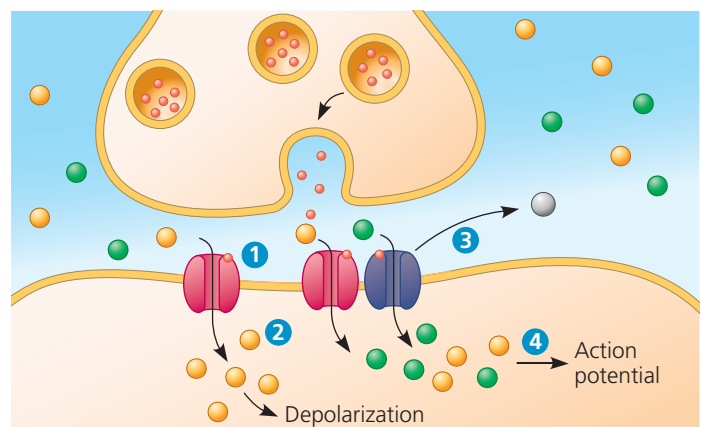
LTP involves two types of glutamate receptors, each named for a molecule—NMDA or AMPA—that can be used to artificially activate that particular receptor. As shown in **Figure 49.21**, the set of receptors present on the postsynaptic membrane changes in response to an active synapse and a depolarizing stimulus. The result is LTP—a stable increase in the size of the postsynaptic potentials at the synapse. Because LTP can last for days or weeks in dissected tissue, it is thought to represent one of the fundamental processes by which memories are stored and learning takes place.



(a) Synapse prior to long-term potentiation (LTP). The NMDA glutamate receptors open in response to glutamate but are blocked by Mg^{2+} .



(b) Establishing LTP. Activity at nearby synapses (not shown) depolarizes the postsynaptic membrane, causing **1** Mg^{2+} release from NMDA receptors. The unblocked receptors respond to glutamate by allowing **2** an influx of Na^{+} and Ca^{2+} . The Ca^{2+} influx triggers **3** insertion of stored AMPA glutamate receptors into the postsynaptic membrane.



(c) Synapse exhibiting LTP. Glutamate release activates **1** AMPA receptors that trigger **2** depolarization. The depolarization unblocks **3** NMDA receptors. Together, the AMPA and NMDA receptors trigger postsynaptic potentials strong enough to initiate **4** action potentials without input from other synapses. Additional mechanisms (not shown) contribute to LTP, including receptor modification by protein kinases.

▲ **Figure 49.21** Long-term potentiation in the brain.

CONCEPT CHECK 49.4

1. Outline two mechanisms by which information flow between two neurons in an adult can increase.
2. Individuals with localized brain damage have been very useful in the study of many brain functions. Why is this unlikely to be true for consciousness?
3. **WHAT IF?** Suppose that a person with damage to the hippocampus is unable to acquire new long-term memories. Why might the acquisition of short-term memories also be impaired?

For suggested answers, see Appendix A.

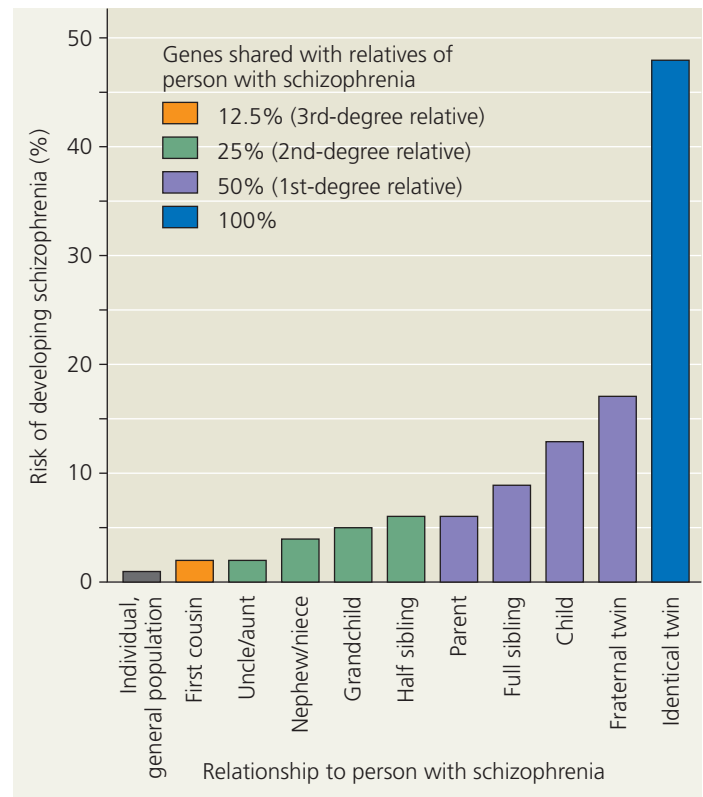
CONCEPT 49.5

Many nervous system disorders can be explained in molecular terms

Disorders of the nervous system, including schizophrenia, depression, drug addiction, Alzheimer's disease, and Parkinson's disease, are a major public health problem. Together, they result in more hospitalizations in the United States than do heart disease or cancer. Until recently, hospitalization was typically the only available treatment, and many affected individuals were institutionalized for the rest of their lives. Today, many disorders that alter mood or behavior can be treated with medication, reducing average hospital stays for these disorders to only a few weeks. Many challenges remain, however, to preventing or treating nervous system disorders, especially for Alzheimer's and other diseases that lead to nervous system degeneration.

Major research efforts are under way to identify genes that cause or contribute to disorders of the nervous system. Identifying such genes offers hope for identifying causes, predicting outcomes, and developing effective treatments. For most nervous system disorders, however, genetic contributions only partially account for which individuals are affected. The other significant contribution to disease comes from environmental factors. Unfortunately, environmental contributions are typically very difficult to identify.

To distinguish between genetic and environmental variables, scientists often carry out family studies. In such studies, researchers track how family members are related genetically, which individuals are affected, and which family members grew up in the same household. These studies are especially informative when one of the affected individuals has either an identical twin or an adopted sibling who is genetically unrelated. The results of family studies indicate that certain nervous system disorders, such as schizophrenia, have a very strong genetic component. However, as shown in **Figure 49.22**, the disease is also subject to environmental influences, since an individual who shares 100% of his or her genes with a schizophrenic twin has only a 48% chance of developing the disorder.



▲ Figure 49.22 Genetic contribution to schizophrenia. First cousins, uncles, and aunts of a person with schizophrenia have twice the risk of unrelated members of the population of developing the disease. The risks for closer relatives are many times greater.

INTERPRET THE DATA What is the likelihood of a person developing schizophrenia if the disorder affects his or her fraternal twin? How would the likelihood change if DNA sequencing revealed that the twins shared the genetic variants that contribute to the disorder?

Schizophrenia

Approximately 1% of the world's population suffers from **schizophrenia**, a severe mental disturbance characterized by psychotic episodes in which patients have a distorted perception of reality. People with schizophrenia typically experience hallucinations (such as “voices” that only they can hear) and delusions (for example, the idea that others are plotting to harm them). Despite the commonly held notion, schizophrenia does not necessarily result in multiple personalities. Rather, the name *schizophrenia* (from the Greek *schizo*, split, and *phren*, mind) refers to the fragmentation of what are normally integrated brain functions.

Two lines of evidence suggest that schizophrenia affects neuronal pathways that use dopamine as a neurotransmitter. First, the drug amphetamine (“speed”), which stimulates dopamine release, can produce the same set of symptoms as schizophrenia. Second, many of the drugs that alleviate the symptoms of schizophrenia block dopamine receptors. Schizophrenia may also alter glutamate signaling: The street drug “angel dust,” or PCP, blocks glutamate receptors and induces strong schizophrenia-like symptoms.

Depression

Depression is a disorder characterized by depressed mood, as well as abnormalities in sleep, appetite, and energy level. Two broad forms of depressive illness are known: major depressive disorder and bipolar disorder. Individuals affected by **major depressive disorder** undergo periods—often lasting many months—during which once enjoyable activities provide no pleasure and provoke no interest. One of the most common nervous system disorders, major depression affects about one in every seven adults at some point, and twice as many women as men.

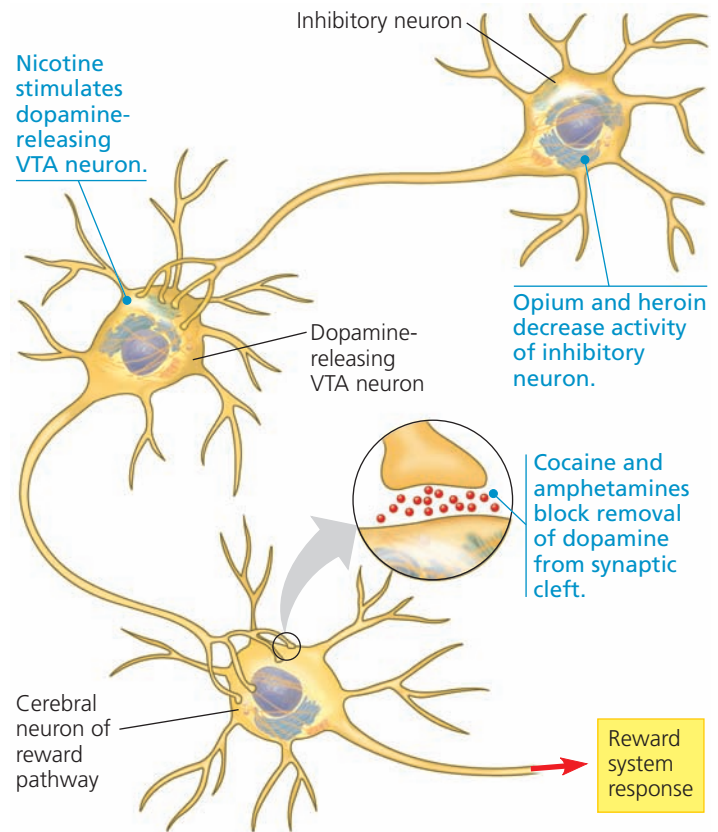
Bipolar disorder, or manic-depressive disorder, involves extreme swings of mood and affects about 1% of the world's population. The manic phase is characterized by high self-esteem, increased energy, a flow of ideas, overtalkativeness, and increased risk taking. In its milder forms, this phase is sometimes associated with great creativity, and some well-known artists, musicians, and literary figures (Vincent Van Gogh, Robert Schumann, Virginia Woolf, and Ernest Hemingway, to name a few) have had very productive periods during manic phases. The depressive phase comes with lowered ability to feel pleasure, loss of motivation, sleep disturbances, and feelings of worthlessness. These symptoms can be so severe that affected individuals attempt suicide.

Major depressive and bipolar disorders are among the nervous system disorders for which effective therapies are available. Many drugs used to treat depressive illness, including fluoxetine (Prozac), increase the activity of biogenic amines in the brain.

The Brain's Reward System and Drug Addiction

Emotions are strongly influenced by a neuronal circuit in the brain called the *reward system*. The reward system provides motivation for activities that enhance survival and reproduction, such as eating in response to hunger, drinking when thirsty, and engaging in sexual activity when aroused. As shown in **Figure 49.23**, inputs to the reward system are received by neurons in a region near the base of the brain called the *ventral tegmental area (VTA)*. When activated, these neurons release dopamine from their synaptic terminals in specific regions of the cerebrum, including the *nucleus accumbens* (see Figure 49.15).

The brain's reward system is dramatically affected by drug addiction, a disorder characterized by compulsive consumption of a drug and loss of control in limiting intake. Addictive drugs, which range from sedatives to stimulants and include alcohol, cocaine, nicotine, and heroin, enhance the activity of the dopamine pathway (see Figure 49.23). As addiction develops, there are also long-lasting changes in the reward circuitry. The result is a craving for the drug independent of any pleasure associated with consumption.



▲ **Figure 49.23** Effects of addictive drugs on the reward system of the mammalian brain. Addictive drugs alter the transmission of signals in the pathway formed by neurons of the ventral tegmental area (VTA), a region near the base of the brain.

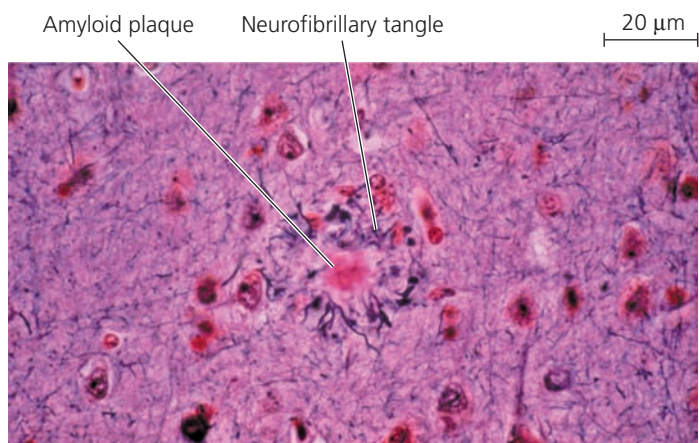
MAKE CONNECTIONS What effect would you expect if you depolarized the neurons in the VTA (see Concept 48.3)? Explain.

Laboratory animals are highly valuable in exploring how the reward system works and how addictive drugs affect its function. For example, Ulrike Heberlein, interviewed at the start of Unit Seven, has shown that the response of fruit flies to alcohol resembles that of humans in several ways, including a loss of coordination and eventually consciousness upon inebriation and a change in responsiveness to alcohol over time. Using fly genetics, she is discovering genes that carry out or control these responses. To study the effects of cocaine or amphetamine in rats, researchers outfit a cage with a dispensing system linked to a lever. In such circumstances, rats exhibit addictive behavior, continuing to self-administer the drug rather than seek food, even to the point of starvation.

As scientists expand their knowledge about the brain's reward system and the various forms of addiction, there is hope that the insights will lead to more effective prevention and treatment.

Alzheimer's Disease

Alzheimer's disease is a mental deterioration, or dementia, characterized by confusion and memory loss. Its incidence



▲ **Figure 49.24** Microscopic signs of Alzheimer's disease. A hallmark of Alzheimer's disease is the presence in brain tissue of neurofibrillary tangles surrounding plaques made of β -amyloid (LM).

is age related, rising from about 10% at age 65 to about 35% at age 85. The disease is progressive, with patients gradually becoming less able to function and eventually needing to be dressed, bathed, and fed by others. Moreover, patients with Alzheimer's disease often lose their ability to recognize people, including their immediate family, and may treat them with suspicion and hostility.

As shown in **Figure 49.24**, examining the brains of individuals who have died of Alzheimer's disease reveals two characteristic features: amyloid plaques and neurofibrillary tangles. There is also often massive shrinkage of brain tissue, reflecting the death of neurons in many areas of the brain, including the hippocampus and cerebral cortex.

The plaques are aggregates of β -amyloid, an insoluble peptide that is cleaved from the extracellular portion of a membrane protein found in neurons. Membrane enzymes, called secretases, catalyze the cleavage, causing β -amyloid to accumulate in plaques outside the neurons. It is these plaques that appear to trigger the death of surrounding neurons.

The neurofibrillary tangles observed in Alzheimer's disease are primarily made up of the tau protein. (This protein is unrelated to the tau mutation that affects circadian rhythm in hamsters.) The tau protein normally helps assemble and maintain microtubules that transport nutrients along axons. In Alzheimer's disease, tau undergoes changes that cause it to bind to itself, resulting in neurofibrillary tangles. There is evidence that changes in tau are associated with the appearance of early-onset Alzheimer's disease, a much less common disorder that affects relatively young individuals.

There is currently no cure for Alzheimer's disease, but an enormous effort has led to the recent development of drugs that are partially effective in relieving some symptoms. Doctors are also beginning to use functional brain imaging to diagnose Alzheimer's disease in patients exhibiting early signs of dementia.

Parkinson's Disease

Symptoms of **Parkinson's disease**, a motor disorder, include muscle tremors, poor balance, a flexed posture, and a shuffling gait. Facial muscles become rigid, limiting the ability of patients to vary their expressions. Cognitive defects may also develop. Like Alzheimer's disease, Parkinson's disease is a progressive brain illness and is more common with advancing age. The incidence of Parkinson's disease is about 1% at age 65 and about 5% at age 85. In the U.S. population, approximately 1 million people are afflicted.

Parkinson's disease involves the death of neurons in the midbrain that normally release dopamine at synapses in the basal nuclei. As with Alzheimer's disease, protein aggregates accumulate. Most cases of Parkinson's disease lack an identifiable cause; however, a rare form of the disease that appears in relatively young adults has a clear genetic basis. Molecular studies of mutations linked to this early-onset Parkinson's disease reveal disruption of genes required for certain mitochondrial functions. Researchers are investigating whether mitochondrial defects also contribute to the more common and later-onset form of the disease.

At present Parkinson's disease can be treated, but not cured. Approaches used to manage the symptoms include brain surgery, deep-brain stimulation, and a dopamine-related drug, L-dopa. Unlike dopamine, L-dopa crosses the blood-brain barrier. Within the brain, the enzyme dopa decarboxylase converts the drug to dopamine, reducing the severity of Parkinson's disease symptoms:



One potential cure is to implant dopamine-secreting neurons, either in the midbrain or in the basal nuclei. Laboratory studies of this strategy show promise: In rats with an experimentally induced condition that mimics Parkinson's disease, implanting dopamine-secreting neurons can lead to a recovery of motor control. Whether this regenerative approach can also work in humans is one of many important questions in modern brain research.

CONCEPT CHECK 49.5

1. Compare Alzheimer's disease and Parkinson's disease.
2. How is dopamine activity related to schizophrenia, drug addiction, and Parkinson's disease?
3. **WHAT IF?** If you could detect early-stage Alzheimer's disease, would you expect to see brain changes that were similar to, although less extensive than, those seen in patients who have died of this disease? Explain.

For suggested answers, see Appendix A.

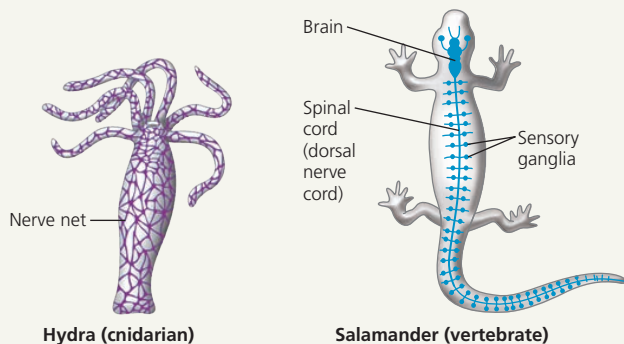
49 Chapter Review

SUMMARY OF KEY CONCEPTS

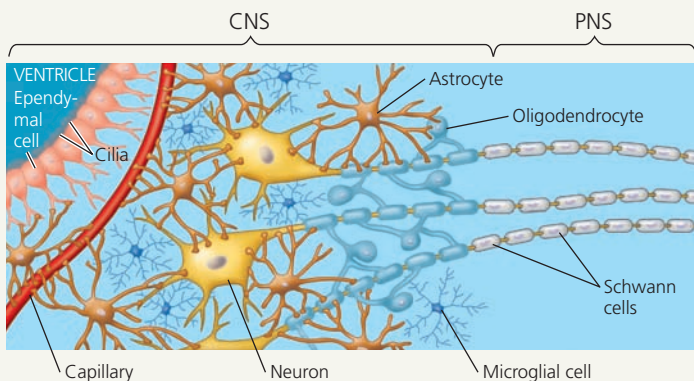
CONCEPT 49.1

Nervous systems consist of circuits of neurons and supporting cells (pp. 1080–1084)

- Invertebrate nervous systems range in complexity from simple nerve nets to highly centralized nervous systems having complicated brains and ventral nerve cords.



- In vertebrates, the **central nervous system (CNS)**, consisting of the brain and the spinal cord, integrates information, while the **nerves** of the **peripheral nervous system (PNS)** transmit sensory and motor signals between the CNS and the rest of the body. The simplest circuits control **reflex** responses, in which sensory input is linked to motor output without involvement of the brain.

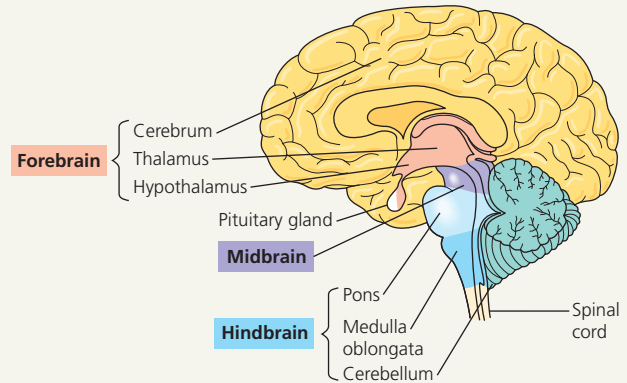


- Afferent neurons carry sensory signals to the CNS. Efferent neurons function in either the **motor system**, which carries signals to skeletal muscles, or the **autonomic nervous system**, which regulates smooth and cardiac muscles. The **sympathetic** and **parasympathetic divisions** of the autonomic nervous system have antagonistic effects on a diverse set of target organs, while the **enteric division** controls the activity of many digestive organs.
- Vertebrate neurons are supported by **glia**, including **astrocytes**, **oligodendrocytes**, and **Schwann cells**. Some glia serve as stem cells that can differentiate into mature neurons.

? How does the circuitry of a reflex facilitate a rapid response?

CONCEPT 49.2

The vertebrate brain is regionally specialized (pp. 1085–1090)



- The cerebrum has two hemispheres, each of which consists of cortical **gray matter** overlying **white matter** and basal nuclei. The basal nuclei are important in planning and learning movements. The **pons** and **medulla oblongata** are relay stations for information traveling between the PNS and the cerebrum. The reticular formation, a network of neurons within the **brainstem**, regulates sleep and arousal. The **cerebellum** helps coordinate motor, perceptual, and cognitive functions. The **thalamus** is the main center through which sensory information passes to the **cerebrum**. The **hypothalamus** regulates homeostasis and basic survival behaviors. Within the hypothalamus, a group of neurons called the **suprachiasmatic nucleus (SCN)** acts as the pacemaker for circadian rhythms. The **amygdala** plays a key role in recognizing and recalling a number of emotions.

? What roles do the midbrain, cerebellum, thalamus, and cerebrum play in vision and responses to visual input?

CONCEPT 49.3

The cerebral cortex controls voluntary movement and cognitive functions (pp. 1090–1093)

- Each side of the **cerebral cortex** has four lobes—frontal, temporal, occipital, and parietal—that contain primary sensory areas and association areas. Association areas integrate information from different sensory areas. Broca's area and Wernicke's area are essential for generating and understanding language. These functions are concentrated in the left **cerebral hemisphere**, as are math and logic operations. The right hemisphere appears to be stronger at pattern recognition and nonverbal thinking.
- In the somatosensory cortex and the motor cortex, neurons are distributed according to the part of the body that generates sensory input or receives motor commands.
- Primates and cetaceans, which are capable of higher cognition, have an extensively convoluted cerebral cortex. In birds, a brain region called the pallium contains clustered nuclei that carry out functions similar to those performed by the cerebral cortex of mammals. Some birds can solve problems and understand abstractions in a manner indicative of higher cognition.

? A patient has trouble with language and has paralysis on one side of the body. Which side would you expect to be paralyzed? Why?

CONCEPT 49.4

Changes in synaptic connections underlie memory and learning (pp. 1093–1096)

- During development, more neurons and synapses form than will exist in the adult. The programmed death of neurons and elimination of synapses in embryos establish the basic structure of the nervous system. In the adult, reshaping of the nervous system can involve the loss or addition of synapses or the strengthening or weakening of signaling at synapses. This capacity for remodeling is termed **neuronal plasticity**. **Short-term memory** relies on temporary links in the hippocampus. In **long-term memory**, these temporary links are replaced by connections within the cerebral cortex.

? Learning multiple languages is typically easier early in childhood than later in life. How does this fit with our understanding of neural development?

CONCEPT 49.5

Many nervous system disorders can be explained in molecular terms (pp. 1096–1098)

- Schizophrenia**, which is characterized by hallucinations, delusions, and other symptoms, affects neuronal pathways that use dopamine as a neurotransmitter. Drugs that increase the activity of biogenic amines in the brain can be used to treat **bipolar disorder** and **major depressive disorder**. The compulsive drug use that characterizes addiction reflects altered activity of the brain's reward system, which normally provides motivation for actions that enhance survival or reproduction.
- Alzheimer's disease** and **Parkinson's disease** are neurodegenerative and typically age related. Alzheimer's disease is a dementia in which neurofibrillary tangles and amyloid plaques form in the brain. Parkinson's disease is a motor disorder caused by the death of dopamine-secreting neurons and associated with the presence of protein aggregates.

? The fact that both amphetamine and PCP have effects similar to the symptoms of schizophrenia suggests a potentially complex basis for this disease. Explain.

TEST YOUR UNDERSTANDING

LEVEL 1: KNOWLEDGE/COMPREHENSION

- Wakefulness is regulated by the reticular formation, which is present in the
 - basal nuclei.
 - brainstem.
 - limbic system.
 - spinal cord.
- Which of the following structures or regions is *incorrectly* paired with its function?
 - limbic system—motor control of speech
 - medulla oblongata—homeostatic control
 - cerebellum—coordination of movement and balance
 - amygdala—emotional memory
- Patients with damage to Wernicke's area have difficulty
 - coordinating limb movement.
 - generating speech.
 - recognizing faces.
 - understanding language.
- The cerebral cortex does *not* play a major role in
 - short-term memory.
 - long-term memory.
 - circadian rhythm.
 - breath holding.

LEVEL 2: APPLICATION/ANALYSIS

- After suffering a stroke, a patient can see objects anywhere in front of him but pays attention only to objects in his right field of vision. When asked to describe these objects, he has difficulty judging their size and distance. What part of the brain was likely damaged by the stroke?
 - the left frontal lobe
 - the right frontal lobe
 - the right parietal lobe
 - the corpus callosum
- Injury localized to the hypothalamus would most likely disrupt
 - regulation of body temperature.
 - short-term memory.
 - executive functions, such as decision making.
 - sorting of sensory information.
- DRAW IT** The reflex that pulls your hand away when you prick your finger on a sharp object relies on a neuronal circuit with two synapses in the spinal cord. (a) Using a circle to represent a cross section of the spinal cord, draw the circuit, labeling the types of neurons, the direction of information flow in each, and the locations of synapses. (b) Draw a simple diagram of the brain indicating where pain would eventually be perceived.

LEVEL 3: SYNTHESIS/EVALUATION

8. EVOLUTION CONNECTION

Scientists often use measures of “higher-order thinking” to assess intelligence in other animals. For example, birds are judged to have sophisticated thought processes because they can use tools and make use of abstract concepts. What problems do you see in defining intelligence in these ways?

9. SCIENTIFIC INQUIRY

Consider an individual who had been fluent in American Sign Language before suffering an injury to his left cerebral hemisphere. After the injury, he could still understand that sign language but could not readily generate sign language that represented his thoughts. What two hypotheses could explain this finding? How might you distinguish between them?

10. SCIENCE, TECHNOLOGY, AND SOCIETY

With increasingly sophisticated methods for scanning brain activity, scientists are developing the ability to detect an individual's particular emotions and thought processes from outside the body. What benefits and problems do you envision when such technology becomes readily available?

11. WRITE ABOUT A THEME: INFORMATION

In a short essay (100–150 words), explain how specification of the adult nervous system by the genome is incomplete.

12. SYNTHESIZE YOUR KNOWLEDGE



Imagine you are standing at a microphone in front of a crowd. Checking your notes, you begin speaking. Using the information in this chapter, describe the series of events in particular regions of the brain that enabled you to say the very first word.

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